



Techno-enviro-socio-economic design and finite set model predictive current control of a grid-connected large-scale hybrid solar/wind energy system: A case study of Sokhna Industrial Zone, Egypt

Mohamed R. Elkadeem^{a,b}, Kotb M. Kotb^c, Mohamed A. Abido^{a,d,e}, Hany M. Hasanien^{f,g,*}, Eman G. Atiya^b, Dhafer Almakhles^h, Mahmoud F. Elmorshedy^{b,h}

^a Interdisciplinary Research Center for Renewable Energy and Power Systems (IRC-REPS), King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia

^b Electrical Power and Machines Engineering Dep., Faculty of Engineering, Tanta University, Tanta 31733, Egypt

^c Interdisciplinary Research Center for Hydrogen and Energy Storage (IRC-HES), King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia

^d Department of Electrical Engineering, King Fahd University of Petroleum and Minerals, Saudi Arabia

^e SDAIA-KFUPM Joint Research Center for Artificial Intelligence, KFUPM, Saudi Arabia

^f Electrical Power and Machines Dep., Faculty of Engineering, Ain Shams University, Cairo 11517, Egypt

^g Electrical Engineering Department, Faculty of Engineering and Technology, Future University in Egypt, Cairo 11835, Egypt

^h Renewable Energy Lab, College of Engineering, Prince Sultan University, Riyadh, Saudi Arabia

ARTICLE INFO

Handling Editor: Dr G Chicco

Keywords:

Hybrid renewable energy system

Grid-connected

Techno-enviro-socio-economic optimization

Model predictive control

ABSTRACT

This article offers a cohesive design optimization and control framework of a large-scale grid-connected battery and battery-less hybrid solar/wind system. Primarily, a techno-enviro-socio-economic design optimization and feasibility analysis were performed for eight distinct energy alternatives. Secondly, a finite-set model predictive current control (FS-MPCC) was proposed to improve both stability and dynamic behavior of the optimal energy alternative under various climatic circumstances. Moreover, finite-set model predictive control is furtherly employed to exploit the maximum permissible power from both solar and wind energies. The proposed framework is verified over a new case-study in Sokhna Industrial Zone in Suez, Egypt. The studied load demand combines both industrial and residential load sectors with a daily consumption of 40,500 kWh and of 4-MW peak. The design optimization results revealed that the grid-connected photovoltaic/wind turbines/converter system has superior sustainable and economic performances with the least net present cost (\$17,371,980) and energy cost (0.0782 \$/kWh). These values were less than those of the base-case system (grid-only) by 17 % and 22 %, respectively. Moreover, the reliability analysis found a tiny and tolerable daily capacity shortage of 5835-kWh despite the five annual grid outages. Besides, around 52 % of the total energy production was acquired from both solar and wind generation systems, which interpreted in a maximum saving in carbon dioxide emissions by 45.8 % compared to the base-case system and offered substantial social benefits in terms of creating nearly 10 jobs. The FS-MPCC results proved a reinforced dynamic behavior for the maximum power tracking profiles of solar and wind systems as well as in the DC-bus voltage profile regarding the over- and undershoot and the steady-state response. Compared to PI controller, the proposed FS-MPCC creates dq-axis voltages and currents with less disturbances and ripples which inject current/voltage to the grid with fewer harmonics and thus increases the generator's lifetime. To sum up, the proposed method features an integrative and compelling case to be used as an efficient tool to implement large-scale renewable energy projects. It could act as a blueprint for energy developers, researchers, investors, and authorities to achieve affordable and sustainable energy access.

* Corresponding author. Electrical Power and Machines Dep., Faculty of Engineering, Ain Shams University, Cairo 11517, Egypt.

E-mail address: hanyhasanien@ieee.org (H.M. Hasanien).

1. Introduction

1.1. Research Context and Motivation

Since the Kyoto Protocol's commencement in 1992, meeting the intensifying energy demand and reducing the onset of global warming using safe, secure, and eco-friendly resources are indispensable [1]. They have been the key goals of world leaders, international organizations, and researchers. As concluded in the UN seventh sustainable development goal (SDG7), achieving a universal access to affordable, reliable, and modern energy public services is essential. Despite of the ongoing progress in the energy sector, around 0.73 billion people since 2020 have been living without access to electricity where the majority of whom are in Africa and Asia (90 million) [2]. It is anticipated that over 0.67 billion people will still be without electricity by 2030. On the other side, the consequences of the Covid-19 pandemic such as lockdowns, disruptions to global supply chains, and restrictive trade measures, have been critical factors in both slowing the progress toward universal energy access and boosting the global electricity demand by more than 6 % [3]. Moreover, the heavy reliance on fossil fuel-based power plants (i.e., coal, natural gas, and oil) largely depleted their natural reserve, increased carbon emissions, caused global warming, and affected the life of organisms.

The aforementioned derives motivated governments, including Egypt, to make significant progress in expanding the deployment of renewable energy sources (RESs), particularly photovoltaic (PV) and wind turbine (WT). The reasons of their selection to address the electricity shortage/crisis and reduce the environmental pollution is that they afford the electricity market needs in modular, sustainable and clean way [4]. Nevertheless, renewables have certain drawbacks such as dependency on uncertain weather conditions, high upfront cost, extended return on investment, large land use footprint, and inability to meet the peak demand without a storage system which obstruct their individual and large-scale applications [5–7]. The previous drawbacks can be overwhelmed throughout integrating one or multiple sources in a so-called hybrid renewable energy system (HRES) [8], which holds great promise for sustainable and long-term electrification and support countries achieving their sustainable goals.

1.2. Literature Review

So far, the development of HRESs has been addressed in many publications from the perspective of system optimization [9], energy management [10], control [11,12], and protection [13]. Several configurations for supplying different end-user customers worldwide considering standalone and/or grid-connected modes have been investigated [14]. Concerning the design optimization problem of HRES, various variables and constraints that rely on the type of system elements, the nature of the weather, the techno-economic parameters, and load patterns make it challenging [15]. The system under-sizing cause

control collapse and a large portion of the unserved load, while over-sizing results in high investments with excessive surplus electricity [16]. Based on the literature, various indicators can be used according to the optimization objectives (i.e., cost, reliability, energy, and emission) [17]. The design optimization problem of HRES is commonly solved using analytical methods, artificial intelligence algorithms, and commercial software tools as comprehensively reviewed in Refs. [17–19]]. Fig. 1 summarizes the classifications of HRES, indicators, and methods used for optimal system design.

To serve the study purpose a prime focus was given to the design optimization and control of grid-connected HRES. For example, the authors in Ref. [20] applied HOMER software for Syiah Kuala University to undertake both the technical and cost feasibilities of a grid-on wind/solar/battery energy supply. In which the energy storage facility is initially added and removed from the simulation to reduce expenses and optimize renewable energy penetration. Their optimization results recommended solar and wind technologies because they decreased the levelized cost of energy (LCOE) of the system to \$0.0446/kWh. Also, the study discovered that RESs could contribute up to 82 % of the required electricity. The optimal sizing of microgrid was also assed in another study performed in Egypt [21]. The authors analyzed and discussed how to optimally size a microgrid to accommodate both centralized and distributed load demands and keep the system's cost and size low. The analytical findings indicate that the design taking centralized load demand into account exhibits the smallest microgrid system size and cost. There is no distinction between the designs considering two bus bars and two centralized load demands and the designs with centralized load demands. The design that considers scattered load demands is more expensive and more extensive.

For a zero-emissions energy solutions, the authors in Ref. [22] proposed a model for cost optimization of microgrid system based on hybrid energy storages for electrifying Ui island in South Korea. The results showed that the PV/WT/battery/fuel cell/hydrogen tank/converter system is the best microgrid configuration over six feasible alternatives with total net present cost (TNPC) of 5,276,069\$ and LCOE of 0.366 \$/kWh. To maintain the electrical demand for a health clinic in Saudi Arabia, the authors in Ref. [23] proposed reliable and cost-effective electrification system combining PV/diesel/battery/fuel cell/electrolyzer. With the 100 % demand shocks created by the Covid-19 virus and with no limitation on fuel availability, the COE of optimal design is reached to 0.130 \$/kWh with more than 40 % renewable fraction. In Ref. [24], HOMER was used to investigate the feasibility of 13 microgrid configurations that consist of a hydrogen system, PV, and WT to replace the electrification of a domestic port from traditional to sustainable while reducing CO₂ emissions. The authors in Ref. [25] investigated the optimal HRES configuration based on biomass and PV, considering solar irradiation, fuel consumption, component efficiencies, and economic convenience as sensitivity variables to affect system performance and cost. In Ref. [26], a hybrid power supply system is created for the Baluchistan Coast. The utility grid is planned to incorporate WT and PV

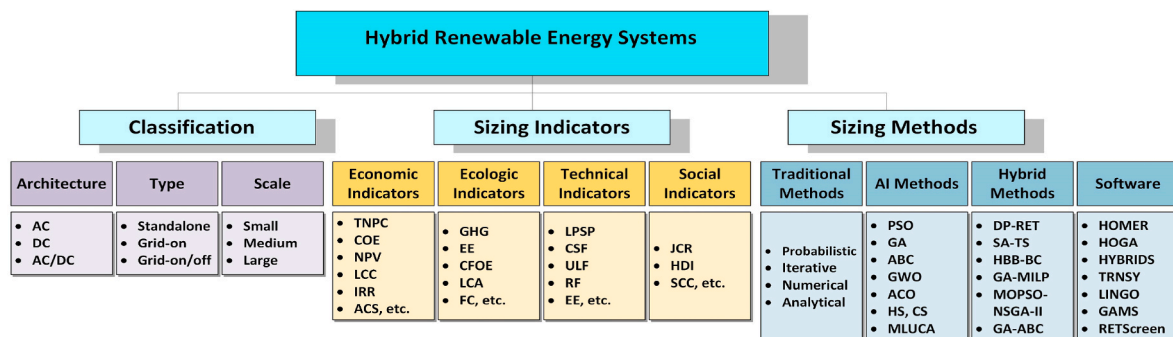


Fig. 1. HRES classifications, optimization objectives, and solution methodologies [17].

installations, reducing the amount owed to the grid. Each power generation case is analyzed based on pollutant gas emissions, net present cost, and average electricity production cost. Moreover, the viability of grid-connected solar PV/diesel/battery to supply industrial parks' demand in various climate zones of Ethiopia under an unplanned grid outage scenario was investigated in Ref. [27]. The simulation results using HOMER software showed the feasibility of the proposed system for all areas with LCOE in the range of 0.044–0.049 \$/kWh. Given the objectives of lower energy costs, net present costs, and renewable proportion, the authors in Ref. [28] offered a techno-economic analysis of grid-connected HRES involving solar, wind, diesel, and storage in tropical regions of the Amazon, Central Asia, and the Mediterranean. Compared to the basic situation with the grid solely, the proposed design achieved reliability prospects of 98 % renewable penetration and 58.2 % emission reduction. In most recent investigations, the authors in Ref. [29], developed an improved search space reduction (ISSR) algorithm for techno-economic optimization of a grid-connected PV/WT/pumped storage microgrid for supplying a household community in Kanyakumari district, India. The obtained results indicate that the proposed microgrid is a viable and scalable solution for meeting the current and future demands, in addition the ISSR yields lower value of LCOE (0.1870 \$/kWh) when compared to the other conventional algorithms. Table 1 summarizes the notable features of the existing studies for the feasibility and optimization of grid-connected HRESs with different loads, system configurations, solution methods, and objectives.

Regarding HRES control, the literature highlights the role of the system control strategy in stabilizing the system and improving the system's dynamic responses and power quality. Each method has its strengths and limitations as summarized in the following section. Among traditional control approaches, Field-oriented control (FOC) method is considered one of the essential techniques to achieve maximum power point tracking (MPPT) of wind generation systems because of its low current total harmonic distortion (THD), constant switching frequency, zero error in steady-state, and robustness to parameter variations [47]. However, this approach needs a modulation stage and responds dynamically and relatively slowly. Another method has been adopted to achieve the same goal of extracting the maximum power, known as direct virtual torque control (DVTC) or direct power control (DPC) [48]. This method depends on hysteresis controllers followed by lookup tables. Although these methods have a faster dynamic response and are more resilient to parameter changes, the switching frequency is variable, and the current harmonics span a wide frequency range. After developments in digital signal processing, advanced control techniques such as intelligent control, sliding mode control, neural networks, and model predictive control (MPC) have been developed to remedy the disadvantages of traditional approaches [49]. The literature analysis show that the most recent developments of the MPC techniques is found in the field of electrical drive systems, including various technical innovations and electric machines as detailed in Refs. [50,51]. Further, the MPC are discussed in several investigations for different power electronics converters [52–54]. Additionally, the MPC is presented for wind generating systems based on different type of generators (PMSG, DFIG, etc) and converters (two-levels, three-levels, etc. [55–61] and also for microgrid and hybrid energy systems [62–69].

Although the MPC is widely used in various applications as mentioned above it has two types for implementation which are continuous control-set MPC (CCS-MPC) and finite control set MPC (FS-MPC). In Refs. [70,71], the CCS-MPC is presented for the PMSMs, taking the system model into account while predicting the system's behavior over a specific prediction horizon. The optimal voltage vector is then chosen by minimizing a particular cost function, and the switching states for the converter are produced using a modulator. A high computational burden is considered the main drawback of the CCS-MPC [72]. On the other side, the FS-MPC strategy was the most promising over the CCS-MPC because it has different merits, such as a restricted number of power converter switching states, simple implementation, fast dynamic

Table 1
Features and limitations of existing studies on the feasibility and design of grid-connected HRES.

Reference	Country name	Load category		Technology type					Algorithm										Design objectives		
				Rural village area	Isolated area	Household	Urban city	desalination	Photovoltaic	Wind turbine	Diesel cell	Fuel cell	Battery	Pumped hydro	Hydropower	Electrolyzer	Hydrogen tank	Biomass	Biogas	Off-grid	On-grid
[28]	China	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[30]	India	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[29]		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[31]	Iran	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[32]		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[33]	Benin	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[34]	China	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[35]		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[36]	Pakistan	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[37]		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[38]	UK	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[39]	Egypt	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[40]		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[41]	Bulgaria	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[42]	South Africa	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[43]	Switzerland	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[44]	Thailand	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[45]	Saudi Arabia	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
[46]	Nigeria	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

responses, the optimal control action used directly or without the use of a modulation stage, can be solved on standard digital signal processors (DSPs), and simple in the handling of nonlinearities and constraints [73, 74]. In the power electronic applications, the deadbeat FS-MPC in Ref. [53] guarantees stable switching frequency, high performance output current waveforms, and strong grid impedance changing characteristics. However, a large number of transformation and space vectors are needed, and this article just focuses on the grid side. Most important, the authors in Ref. [54] proposed eliminating the instability problem caused by the constant power load (CPL) in the DC-DC buck converter applied in DC microgrids.

Meanwhile, for the wind generating system, the authors in Ref. [55] provided a direct model predictive control (DMPC) method for a grid-connected PMSG wind energy system with an LCL filter. The DMPC includes the LCL filter values in the modeling and the control processes. Also, to select the appropriate weighting factors for the switching frequency control, which results in a relatively small range of switching frequencies, a heuristic technique is used. Nevertheless, this DMPC method only concentrates on the grid side converter and uses a computationally intensive heuristic technique to choose the weighting factor. Another developed study [56] proposed FS-MPC for the direct power control for the grid side converter without any attention to machine side converter, which is essential in extracting the maximum power from the wind. In Ref. [57], the wind generating system using PMSG based on a novel nonlinear model-based predictive controller without speed/position sensors is developed. However, there is little explanation of the wind power's MPPT, and choosing the appropriate values for three weighting in the cost function take more work. Ref. [58] developed a variable-speed operation technique suitable for WT-powered RO systems using an optimized operational strategy integrated with MPC considering wind power variability. Moreover, a predictive current controller for grid-connected wind energy systems is presented in Ref. [75]. This implementation uses a boost converter after a non-controlled rectifier to regulate the maximum wind power. The same concept was adopted in Ref. [59] but with upgraded type of converter where three-level (3L) is used instead of two-level (2L). The power conversion system consists of an uncontrolled rectifier, 3L boost converter and 3L neutral-point clamped inverter. This paper investigates a predictive control scheme to operate the converter system with fixed switching frequency, high-speed tracking, and accurate locking in steady state. However, these systems result in strong torque pulsations on the generator, require large inductor in the boost converter, and need additional PI control to produce the reference torque. Another study has been performed by replacing the boost converter with a regulated rectifier, similar to that shown in Ref. [60], while paying little heed to the machine side converter control. On the other hand, the authors in Ref. [61] applied a control technique based on the MPC and a back to back converter. Meanwhile, the dc bus voltage response is detorting because the system is only paying attention to the wind generating system and ignoring the PV system.

In microgrid and hybrid energy systems, the implementation of MPC facilitates the multivariable control of power electronic systems while accommodating physical constraints without requiring a cascaded structure [76]. It provides flexibility to model system constraints and nonlinearities. In Ref. [62], distributed generation, a low-voltage microgrid, energy storage, and load are coupled to a medium-voltage grid to form a microgrid cluster system. In this work, the MPC is suggested to solve the problem of the three-phase imbalanced control in this cluster system. Meanwhile, [63], describes a two-stage optimal structure for the online dispatch in the DC microgrid, where the first stage models the power coordination to get the schedule plans for the grid, the charging station, and the storage unit. To address issues of uncertainty, a combined robust and stochastic MPC strategy using several granular models is provided. The second stage involves determining the charging power for each electric vehicle using the charging station allocation model. In Ref. [64], this provides a paradigm for cooperative control for

a networked group of microgrids with multiple intelligent greenhouses. The main goal is to develop a coordinated optimization framework integrated into the MPC scheme to efficiently manage the exchange of power flows and govern the operation of the clustered microgrids while assuring excellent service quality. In Ref. [65], the minimum operating cost and the efficient use of the RES for cotton farm microgrid is based on the MPC for both grid-connected and off-grid. Meanwhile, Refs. [66,68] are proposed the MPC for the management of energy resources in a residential microgrid. In order to decrease the collective economic cost (including peak power cost and spot-market cost) of the residential coalition then distribute the obtained profits to all residents. Further, the MPC is discussed for the DC microgrid in the device-level control [67, 69] to control the output from every modules of the hybrid microgrid. Additionally, the proposed MPC in Ref. [69] is used to achieve an efficient power management.

1.3. Research Gaps and Contributions

Based on the previous survey, it can be realized that the enthusiastic development of RESs as an alternative solution to fossil fuels is the prevalent track of fighting to achieve worldwide carbon neutrality and meet sustainable development goals 7 and 13 [77]. Nevertheless, the large-scale application of RESs in power systems is yet in the precursory phase of global energy evolution, particularly in developing countries. Giving a closer look, the incorporated technologies and implementation techniques for large-scale renewable projects are not even adequately deliberated [78]. The main key contributions of our study in view of the current scientific gaps can be summarized as follows:

- Most existing literature that focuses either on the design [22–46] or control [62–69] issues of HRESs, particularly small-scale systems. Besides, less attention has been paid to in-depth analyses of the existing challenges in large-scale RESs applications for carbon impartiality evolution [79]. In this study, a four-phase integrated and coherent research method was developed they simultaneously guarantees the optimal design and efficient control of MW-scale on-grid HRES. This method broadens the theoretical knowledge and could be considered as a preliminary efforts for practical application of large-scale HRESs.
- As the stochastic and intermittent nature of both solar and wind energies in time and value expedites distinct challenges in coordinating energy generation and consumption in single use applications; consequently, the sensible functioning efficacy of such RESs would be vigorously demoralized [80,81]. The proposed HRES couples renewable energy conversion (PV and WT) with battery energy storage and the utility grid. The use of solar and wind resources engenders a complementarity behavior reducing their inherent and variable characteristics that would ameliorate the system's reliability, operability, and investment opportunity.
- To the best of the authors' knowledge, the scholarly studies on the design of HRESs were extensive but they were devoted to using economic measures (e.g., Refs. [20,21,29,34,36,39,45]). This shortcoming could obscure the right decision outcomes of the project portfolios and jeopardises the project sustainability. Herein, a thorough modeling was developed in which the simulation approach for techno-enviro-socio-economic feasibility and optimization with was implemented with the aid of HOMER Pro software. Eight different energy supply alternatives are optimally designed and compared in view of energy (ExEL, CShF), emissions (RenF, CO₂), social (JCF), and financial (TNPC, LCOE, CAPEX, OPEX) metrics.
- Compared to other investigations except [62] that mostly apply MPC to control the WT system [55–61] or energy management of hybrid systems [63–69], we proposed FS-MPCC for the machine side converter to extract the maximum power from the WT generating system and for the boost converter combined with the PV system to maximize the extracted energy and hence increase the efficiency and also

improve the entire performance of the proposed hybrid system. The MPC for the PV features the dependency on the combination of the incremental conduction method with the FS-MPCC. The MPC for the WT is based on current control with the finite set concept because of getting fast tracking, simple implementation, robust performance, and removing the weighting factor from the cost function.

- The proposed FS-MPCC aims to strengthen to strengthen dynamic performance and increase the stability of the DC bus voltage of the hybrid systems against the dynamic fluctuations of weather data, thus facilitating the implementation large-scale renewable energy projects. Moreover, increasing the lifetime of the WT system as the developed torque ripples are decreased, and the injected waveforms of the current become smooth as the DC-bus voltage is stable. For more convening, the performance outcomes of the proposed FS-MPCC and the traditional PI controller are adequately compared and analyzed.
- The large space of the reviewed literature has investigated the application of on-grid HRESs in supplying rural villages [30,31,36,37], urban cities [33,43], remote islands/coastal regions [26,32,44], residential households [28,29,32,44,46], desalination plants [38,40,45], university buildings [20,21]. Hence, more attention should be given to measuring and assessing the applicability of HRES for industrial facilities. To do this, we draw on a new representative case study of a group of industrial and residential loads of 14777.692 MWh/year in Sokhna Industrial Zone, Suez city, Egypt to test and verify the method effectiveness. All essential parameters such as hourly load distribution pattern, meteorological resources, and component costs, and technical specs are considered. Accordingly, this research effort offers new insights and a useful paradigm, which can be used by investors, policymakers and researchers whose showing interest in RESs implementation, especially for green electrification of the industrial sector.

1.4. Paper Organization

The leftover of this paper is structured as follows: Section 2 provides the research materials and methods details, including the design optimization approach and control techniques. Section 3 presents the case study information, including resources, location significance, and load

assessment. The results are demonstrated and discussed in Section 4. Finally, the conclusions and future recommendations are presented in section 5.

2. Materials and Methods

This study presents an integrated optimization and control approach for a large-scale solar/wind/battery/grid hybrid power system in a coastline city in Egypt. The system is used to supply a typical residential community and industrial loads. The combination of solar and wind renewable sources is adopted given their good complementary characteristics and provide a sustainable and clean supply. If necessary, the energy storage is equipped to maintain the load demand in case of renewables deficits and reduce the energy purchase from the grid. Fig. 2 illustrates a schematic diagram of the studied system. The components of the hybrid renewable energy system are a PMSG driven by a wind turbine, photovoltaic panels, a three-phase controlled rectifier, a boost converter, a three-phase voltage source inverter (VSI), a filter, and the utility grid. The additional material details the mathematical modeling and system elements' specifications.

The framework of the research method is demonstrated in Fig. 3. First, the feasibility and design optimization of different power system configurations is performed using the HOMER tool (phase-1, phase-2, and phase-3). This is to determine the system elements' optimal capacities at low cost subject to demand and supply balance and other system constraints. In the second stage (phase-4), a proposed MPC control method is applied to manage the injected active and reactive powers into the grid with low complexity and smooth dynamic response.

2.1. Proposed Feasibility and Design Optimization Method

In the first stage of the research method, we used HOMER Pro software [82] to conduct techno-enviro-socio-economic feasibility and design optimization of the proposed hybrid power system under the real conditions of local renewable and load resources of the investigated location. According to Mazzeo et al. [9] and Lian et al. [17], HOMER has been used in countless studies as the trusted global standard in hybrid power system modeling, simulation, and optimization in all sectors. This tool features a depth of analysis with an outstanding balance between

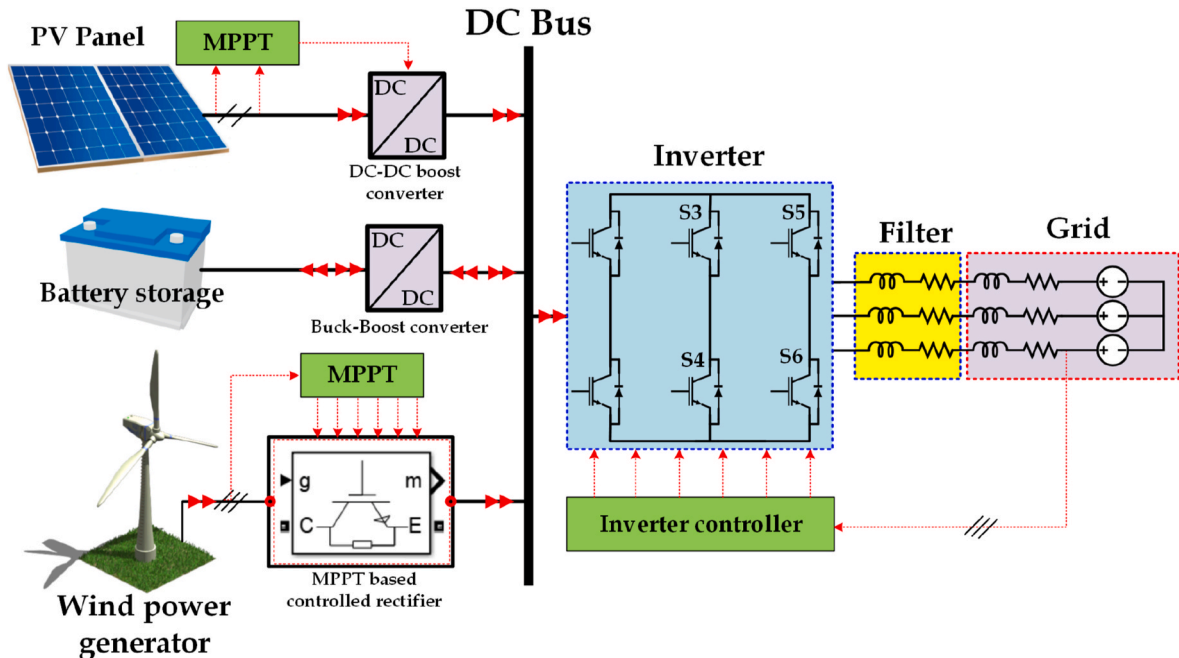


Fig. 2. Schematic diagram of the proposed hybrid energy system.

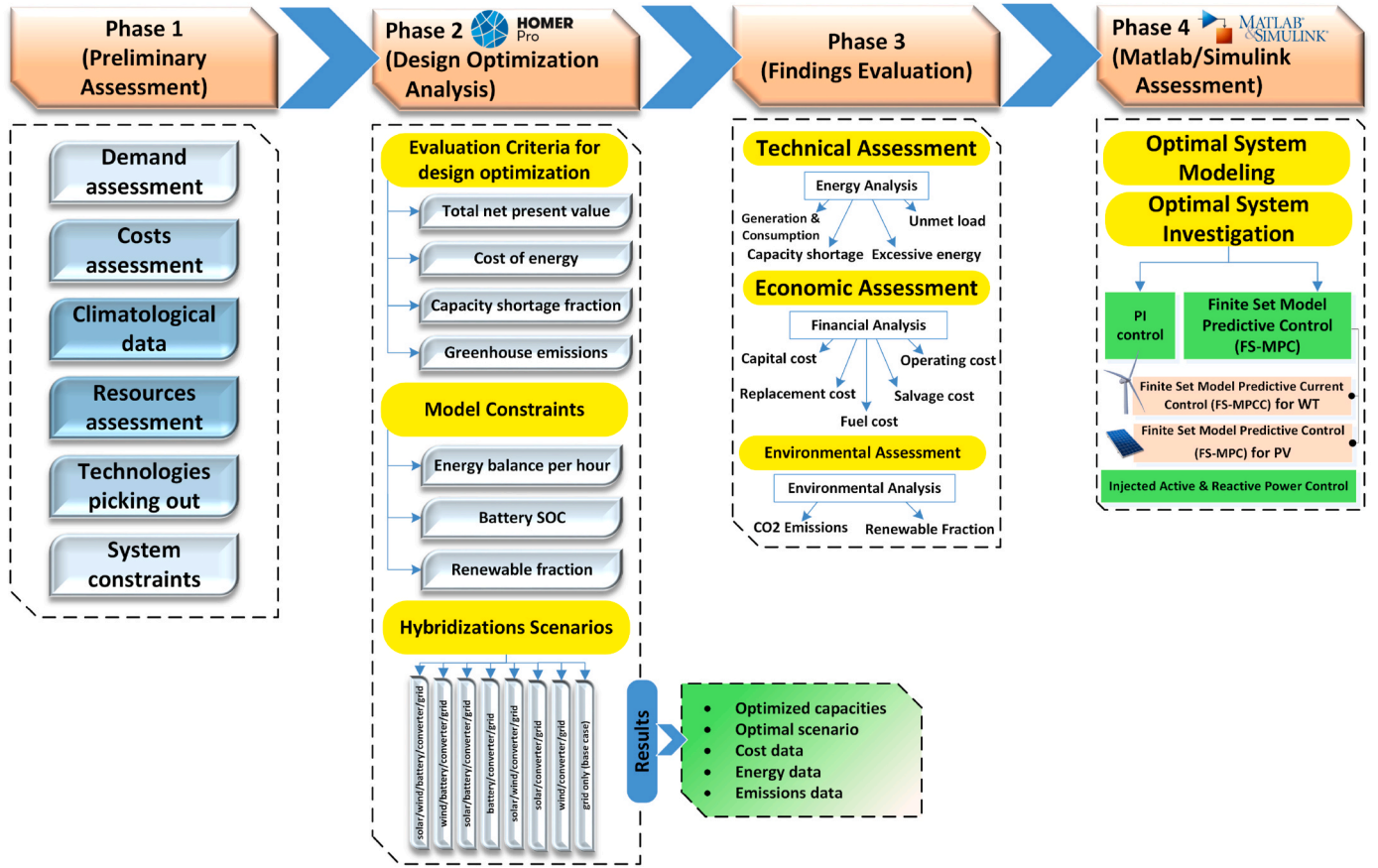


Fig. 3. Representation of the research framework.

thoroughness, simplicity, and flexibility [83]. Fig. 4 illustrates the techno-economic feasibility and design optimization approach. Firstly, a pro-practicability appraisal of the weather of the project area, economic considerations, technical requirements, and energy demand is performed. Using the HOMER algorithm, we attempt to simulate and optimize the operation of all possible hybrid configurations of the proposed power system for an entire year in time steps of 1 min. The optimal values of the decision variables (capacities of solar PV and converter, number of wind turbines and batteries and grid purchase power) are calculated in each loop of the algorithm. The process is repeated till the precision convergence of the optimization variable (i.e., the objective function, TNPC) is accomplished, and all alternatives are simulated. After that, the optimized systems are sorted, and the winning design is identified at the least cost. It is worth stating that two approaches of energy dispatch are considered while looking for the optimal system design: cycle charging and load following. Ramesh and Saini [84] studied both strategies' implementation philosophy and discuss both advantages and disadvantages. The one that achieved the least system cost is selected with each configuration.

2.1.1. Total net present cost

The total net present cost is defined as the total costs of the HRES invested over its lifetime, minus all present value of all revenue it receives over its lifetime. This cost type can be computed by totaling the discounted cash flows per year over the project's life cycle. As a comprehensive method to assess and compare various HRES alternatives that differ with respect to initial costs and operating costs, the TNPC is considered as the main objective function to be minimized, as expressed in Eq. (1) [8]:

$$\begin{aligned} \text{Min TNPC} = & \text{CAPEX} + \sum_{n=1}^N \text{OPEX}_n \cdot \frac{1}{(1+d_r)^n} + \sum_{n=1}^N \text{FLC}_n \cdot \frac{1}{(1+d_r)^n} \\ & + \sum_{n=1}^N \text{RPC}_n \cdot \frac{1}{(1+d_r)^n} - \sum_{n=1}^N \text{SVC}_n \cdot \frac{1}{(1+d_r)^n} \text{ where } d_r = \frac{d_n - d_f}{1 + d_f} \end{aligned} \quad (1)$$

where CAPEX is the capital expenditure of the HRES system (\$), and OPEX_n is the operational expenditure of the HRES system in the n th year (\$). FLC_n , SVC_n , RPC_n is the fuel, replacement, and salvage costs of the system in the n th year (\$). Besides, d_r is the real discount rate (%), N is the project lifetime (yr), d_n is the nominal discount rate (%), and d_f is the anticipated inflation rate (%).

2.1.2. Levelized cost of energy

The levelized cost of energy represents the minor cost of selling electrical energy at a break-even price for the HRES lifetime. It is expressed as the average cost per kWh of useful electrical energy generated by the microgrid system as follows [85]:

$$\text{LCOE} = \frac{C_{\text{tot,an}}}{E_{l,s}} \quad (2)$$

$$C_{\text{tot,an}} = \text{TNPC} \cdot \text{CRF}(d_r, N) \quad (3)$$

$$\text{CRF}(d_r, N) = \frac{d_r \cdot (1 + d_r)^N}{(1 + d_r)^N - 1} \quad (4)$$

where $C_{\text{tot,an}}$ is the total annual cost (\$/year) of the HRES, and E_g is the yearly total electricity production (kWh/yr), $E_{l,s}$ is the yearly total electrical load served (kWh/yr), and CRF is the capital recovery factor.

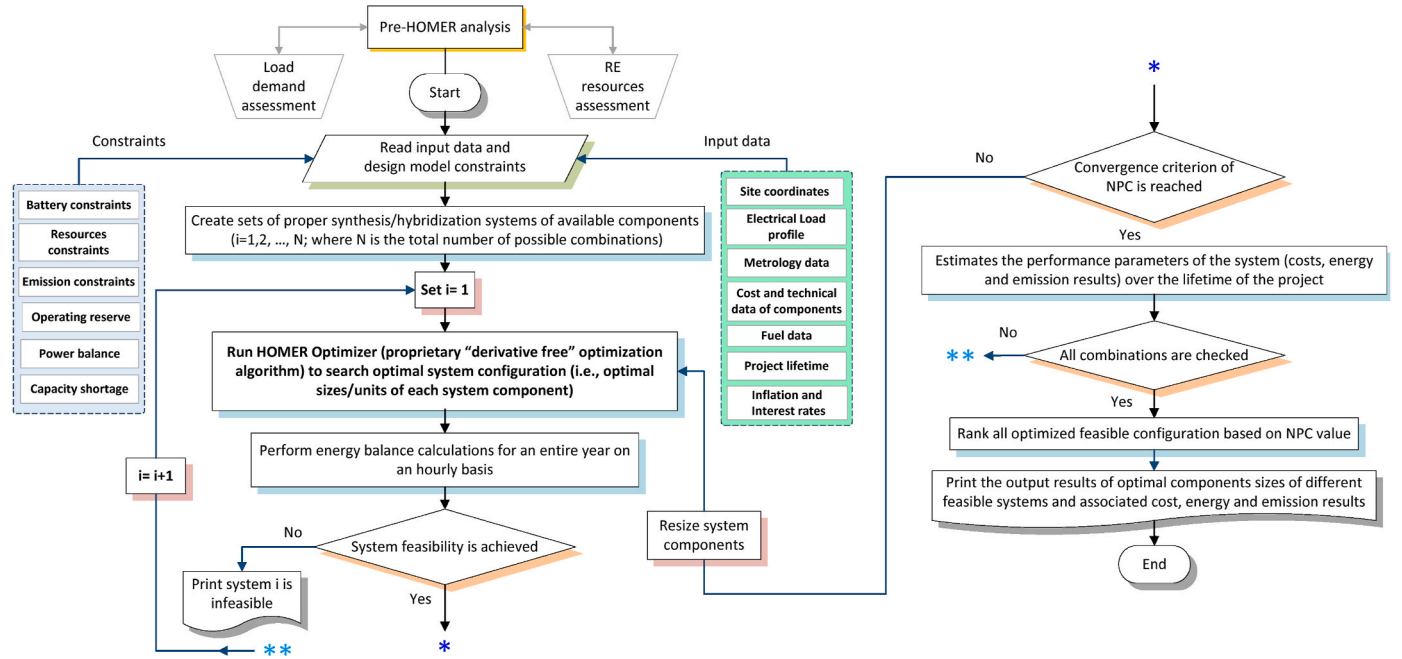


Fig. 4. Flowchart of the employed method for feasibility and optimization of HRES.

2.1.3. Capacity shortage fraction

The capacity shortage fraction describes the possibility of any energy discrepancy in a system at a specific time. As in Eq. (7) [15], it can be expressed as the absolute energy scarcity concerning the yearly electric load. The capacity shortage fraction (CSF) is utilized as a consistency gauge to examine and evaluate service endurance. The values of 0 % and 1 % CSF denotes that the load will never be served and the load will be fully fulfilled.

$$CSF = \frac{E_l - E_g}{E_l} \quad (5)$$

2.1.4. Carbon dioxide emissions

The amount of carbon dioxide emissions is a measure of the HRES environmental performance. In this study, the CO₂ value differed from one alternative to another according to the amount of energy imported from the power grid and can be calculated using Eq. (6).

$$CO_2 = e_f \cdot E_{Grid,im} \quad (6)$$

where $E_{Grid,im}$ is the yearly total energy imported/purchased from the utility power grid (kWh/year), and e_f is grid emission factor for CO₂ pollutant (0.632 g/kWh).

2.1.5. Job creation factor

The job creation factor shows the number of jobs derived from manufacturing, construction & installation, operation & maintenance, and of decommissioning of energy system technologies [86]. It represents an essential factor to assess the social benefits of each alternative. The calculation of job creation factor (JCF) of a certain system is usually done through the use of employment factors of the configuration technologies as expressed in Eq. (7) [87,88].

$$JCF = EF_{PV} \cdot P_{PV} + EF_{WT} \cdot P_{WT} + EF_{Grid} \cdot E_{Grid,im} + EF_{BAT} \cdot E_{BAT} \quad (7)$$

where EF_{PV} and EF_{WT} are the employment factor the utility-scale PV (2.7 jobs/MWp) and onshore WT (1.1 jobs/MW), respectively. EF_{Grid} and EF_{BAT} are the employment factor for the energy purchased from the

utility grid (e.g., gas power plant) (0.11 jobs/GWh/year) and battery storage (0.01 jobs/GWh rated capacity), respectively.

3. Proposed Finite-set Model Predictive Current Control Technique

The main target of the suggested FS-MPCC technique is to extract the maximum power from the WT and PV generation systems using the controlled rectifier and the boost converter, respectively. Meanwhile, the active and reactive injected power to the grid is based on the sinusoidal pulse width modulation (SPWM) to obtain fixed switching frequency with high values; hence the waveforms of the injected current can be improved. The conventional proportional-integral (PI) control technique is also modeled for WT for comparative analysis and robustness check of the FS-MPC.

3.1. Conventional PI Control Technique for WT Generation System

As previously explained, the controlled rectifier is employed instead of the boost converter and the uncontrolled rectifier. Hence the control method is entirely different, where six pulses are required rather than one pulse. The dq -axis currents are employed for control based on the dynamic model of the PMSG, where each can achieve a different goal, as expressed in Eq. (8). The d -axis is utilized to cut losses while the q -axis reference current is used to extract the most power. The optimal wind power, P_{opt} , is used to determine the reference q -axis reference current, i_{qs}^* , which can be provided by Eq. (9), where the constant 0.24 is based on the wind turbine constant values [89].

$$i_{qs}^* = \frac{4 \cdot P_{opt}}{3 \cdot p \cdot \lambda_m \cdot \omega_T} \quad (8)$$

$$P_{opt} = 0.24 \cdot \rho \cdot A \cdot V_w^3 \quad (9)$$

To minimize the losses, the reference point of the d -axis current is set to zero. By deciding the set points of the dq -axis currents, the real values can track these set points by using two PI controllers. After that, the

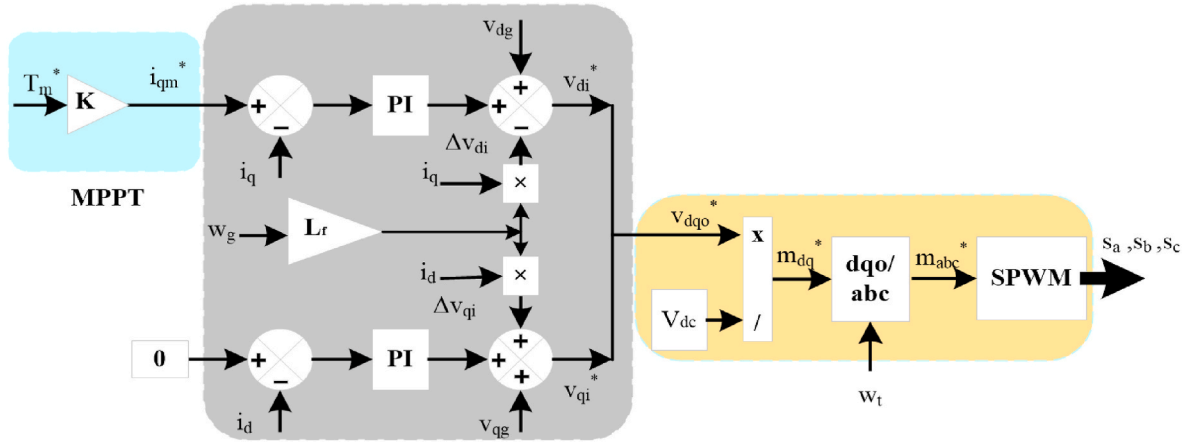


Fig. 5. Conventional PI control method to extract the MPP from the WT generation system.

error signals between the currents are the input to these two PI controllers. To get a complete decoupling, compensation is used. Finally, the pulses are generated based on the compensated values and after transformed to abc references and the SPWM method. Fig. 5 shows this conventional PI control method.

3.2. The Proposed FS-MPCC for the WT Generation System

The block diagram of FS-MPCC applied for WT is shown in Fig. 6. The FS-MPCC is utilized in place of the conventional PI control method to get a faster transient and improved dynamic response. In addition, the PI is removed; hence, the tuning effort for the four gains is avoided. Further, the pulses of the six switches are generated directly based on the lower cost function value. To optimize the efficiency of this control algorithm, it can be split into three critical steps; measuring the required states,

prediction, and optimization, defined by the cost function as presented below: -

3.2.1. Measuring the required states

In this step, the three-phase electrical current produced is first accurately measured. Following this, these measurements are transformed into dq axis values. Additionally, this procedure involves the precise measurement of the rotational speed of the PMSG. This comprehensive approach ensures both the electrical output and the mechanical operation of the generator are closely monitored and analyzed.

3.2.2. Prediction step

In this second step, the prediction states are dq-axis currents, where the first-order Euler method is used to predict the values in the next step.

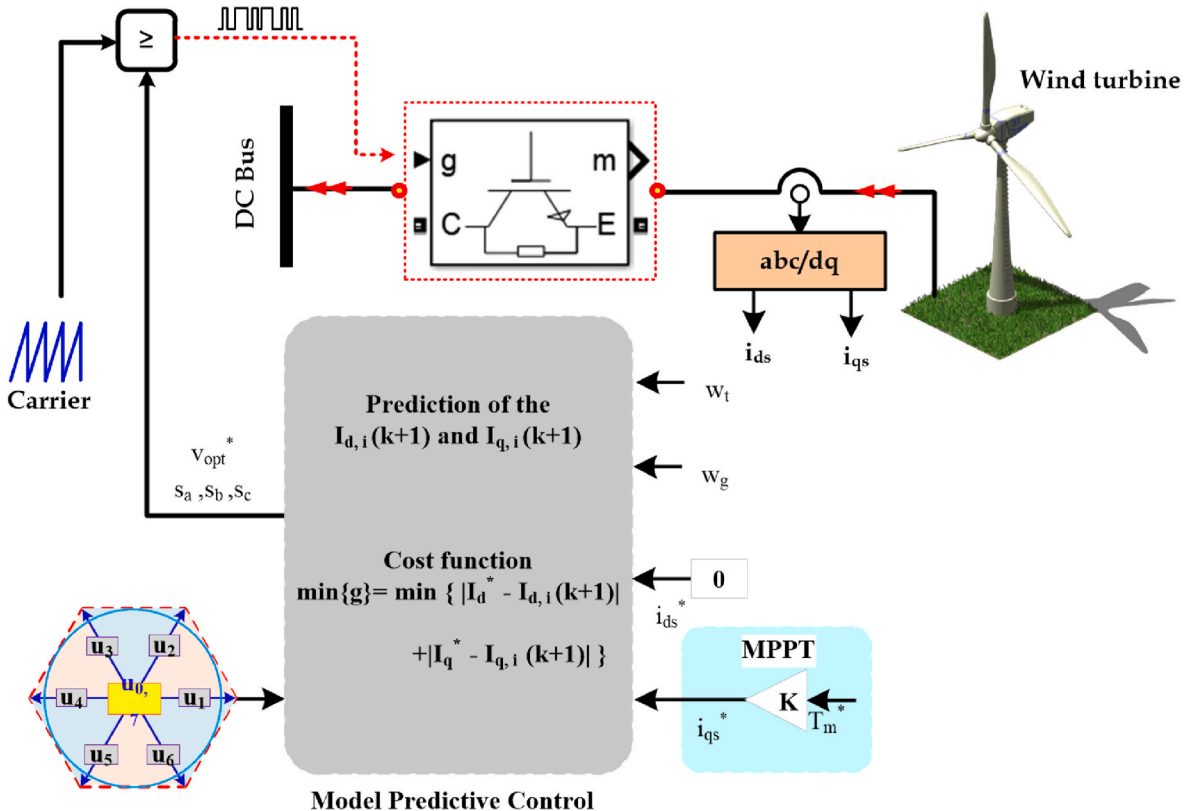


Fig. 6. Block diagram of the suggested FS-MPCC to extract the maximum power from the WT.

These states are selected to be used in the third step, which is the cost function; hence, there is no need to weigh factors. The mathematical model of the PMSG is used to get these values as expressed by Eqs (10) and (11).

$$I_{q,i}^{k+1} = I_q^k + T_s \cdot \left(\frac{u_{q,i}^k - I_q^k \cdot R_s - \omega_e \cdot \psi}{L_q} + \omega_e \cdot I_d^k \right) \quad (10)$$

$$I_{d,i}^{k+1} = I_d^k + T_s \cdot \left(\frac{u_{d,i}^k - I_d^k \cdot R_s}{L_d} + \omega_e \cdot I_q^k \right) \quad (11)$$

where I_d^k and I_q^k are the dq-axis measured currents, ω_e is the electrical rotating speed, and ψ is the permanent magnet flux linkage.

3.2.3. Adopted cost function

The reference set point is tracked using the adopted cost function in this case, and the weighting factor is eliminated because of the same

units of the items used in this cost function. The design of the proposed cost function, g , is given by:

$$g = |I_d^* - I_{d,i}^{k+1}|^2 + |I_q^* - I_{q,i}^{k+1}|^2 + I \quad (12)$$

where I is a constraint used to conserve the machine from both over current and the selection of the wrong switching vector.

3.3. The Proposed FS-MPCC for PV

In this article, FS-MPC is used along with the IC to extract the MPP from the PV panels. The reference inductor current (i.e., PV optimum current, $i_{ref,pv}$) of the boost converter is generated from the IC method. The flowchart of generating this current is shown in Fig. 7. After getting this optimum current, the following cost function in Eq. (13) is used to generate the switching pulse for the boost converter. The block diagram of the suggested FS-MPC to extract the MPP from the PVS is shown in

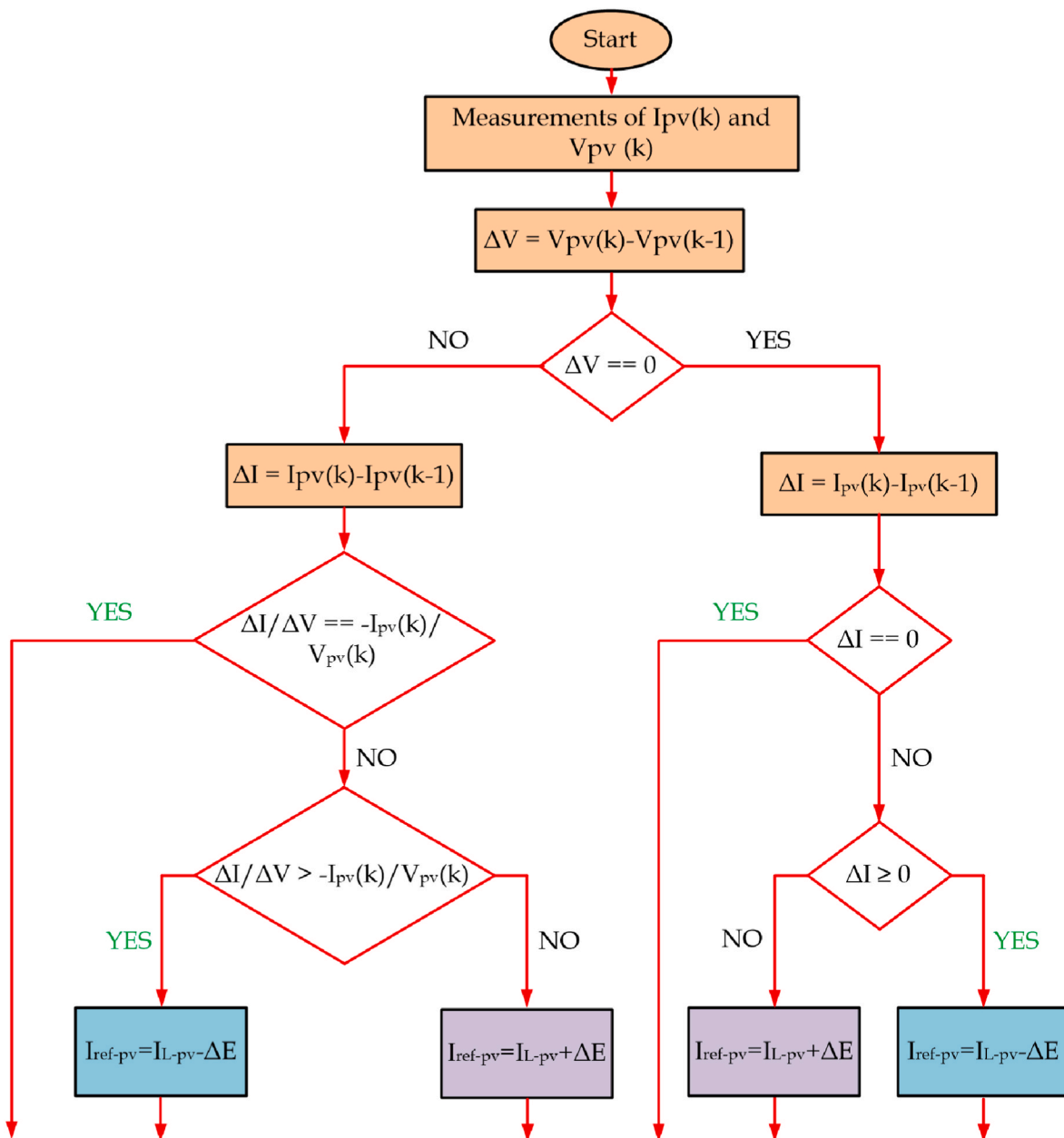


Fig. 7. Flowchart of generating the reference PV current based on the IC method.

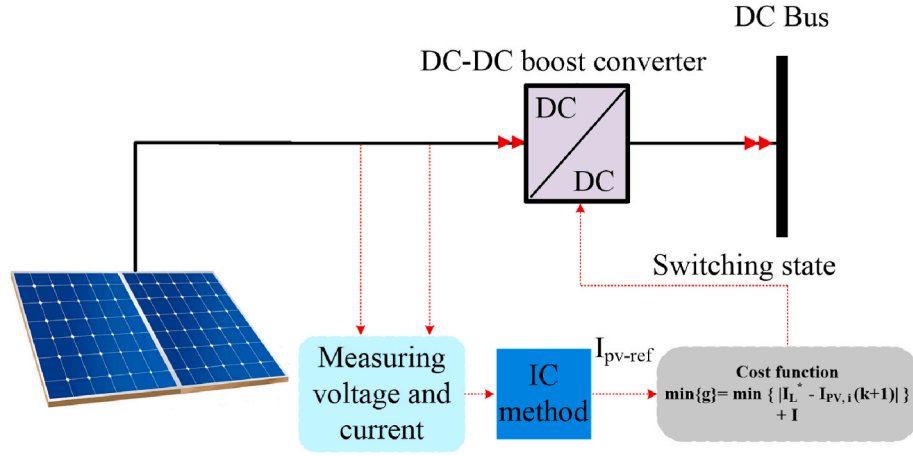


Fig. 8. The FS-MPC control technique to extract the maximum power from the PV system.

Fig. 8.

$$g = |I_L^{k+1} - i_{ref,PV}| + I \quad (13)$$

where I_L^{k+1} is the predicted current of the inductor and I the penalty function to avoid the damage of the PV, which can be calculated by Eqs. (14) and (15), respectively.

$$I_L^{k+1} = I_L^k + \frac{T_s}{V_{boost}} (V_{PV-output} + V_{DC-bus} \bullet (X-1)) \quad (14)$$

$$I = \begin{cases} \infty & \text{if } I_{SC}^{k+1} > I_{sc} \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

where X is the switching pulse, and it is 1 or 0, and T_s is the sampling time.

3.4. Active and Reactive Power Injected into the Grid

This section analyses the control method for controlling the active and reactive power injected into the grid. During this control, the DC-bus voltage is regulated to maintain constant at the specified level. Hence, the DC-bus voltage control and the phase-locked loop (PLL) are discussed first to facilitate the analysis of the grid side control (i.e., active and reactive power control).

By regulating the d-axis output current, the DC-bus voltage is maintained constant. The input-output power balance and the assumption that the system is lossless illustrate this, as stated by Eq. (16).

$$v_{DC-bus} \cdot C \cdot \frac{dv_{DC-bus}}{dt} = \frac{3}{2} (v_{dg} \cdot i_{dg} + v_{qg} \cdot i_{qg}) \quad (16)$$

When the voltage-oriented control concept is adopted, the whole voltage magnitude is oriented in the direction of the d-axis $v_{dg} = \hat{V}_{\phi 1}$, hence Eq. (17) can be updated as in Eq. (18).

$$v_{DC-bus} \cdot C \cdot \frac{dv_{DC-bus}}{dt} = \frac{3}{2} v_{dg} \cdot i_{dg} \quad (17)$$

After linearization around the equilibrium point, Eq. (17) can be represented in a new form in Eq. (18) where $\hat{v}_{dg}$ and $\hat{i}_{dg}$ are the perturbed values.

$$V_{DC-bus} \cdot C \cdot \frac{dv_{DC-bus}}{dt} = \frac{3}{2} \hat{v}_{dg} \cdot I_{dg} + \frac{3}{2} V_{dg} \cdot \hat{i}_{dg} + \frac{3}{2} V_{dg} \cdot I_{dg} + \frac{3}{2} \hat{v}_{dg} \cdot \hat{i}_{dg} \quad (18)$$

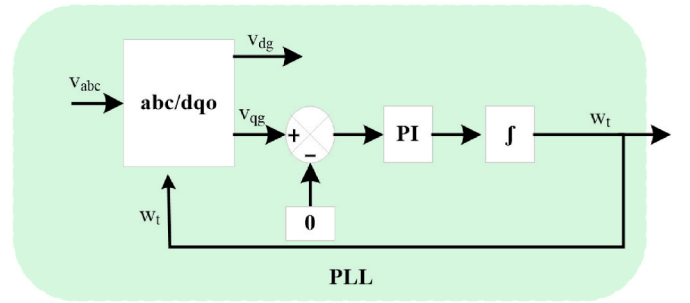


Fig. 9. Block diagram showing the phase locked loop control.

Eq. (18) is then modified after ignoring the higher-order; the steady-state terms and the final form are expressed in Eq. (19).

$$V_{DC-bus} \cdot C \cdot \frac{dv_{DC-bus}}{dt} = \frac{3}{2} \hat{v}_{dg} \cdot I_{dg} + \frac{3}{2} V_{dg} \cdot \hat{i}_{dg} \quad (19)$$

It can be noticed that the DC-bus voltage is affected by the value of the d-axis current; hence a PI controller can be used to adjust this value.

The extraction of the grid voltage vector angle is considered the PLL's primary goal, as indicated in Fig. 9, to properly align the d-axis with the voltage vector in the VOC [90]. The value of the d-axis voltage is zero, $v_{dg} = 0$, the voltage on the q-axis is stated by Eq. (20) because the PLL's action lines up the d-axis with the voltage vector where w_t is the angle of the d-axis concerning α -axis and ωt_o the voltage angle with respect to α -axis.

$$v_{qg} = \hat{v}_{\phi 1} \cdot \sin(\omega t_o - \omega t) \quad (20)$$

Aided by the previous analysis, the control of the active and the reactive power values can be achieved. To relate the relation between the output signal from the PI controller of DC-bus voltage and the grid side current control, which is used in regulating the active and the reactive power, the inverter output voltages are developed in the dq -axis as shown in Eqs. (21) and (22).

$$v_{di} = (R_f + L_f \cdot p) \cdot i_{di} - \omega_g \cdot L_f \cdot i_{qi} + v_{dg} \quad (21)$$

$$v_{qi} = (R_f + L_f \cdot p) \cdot i_{qi} - \omega_g \cdot L_f \cdot i_{di} + v_{qg} \quad (22)$$

Where R_f , L_f are the resistance and inductance of the filter between the

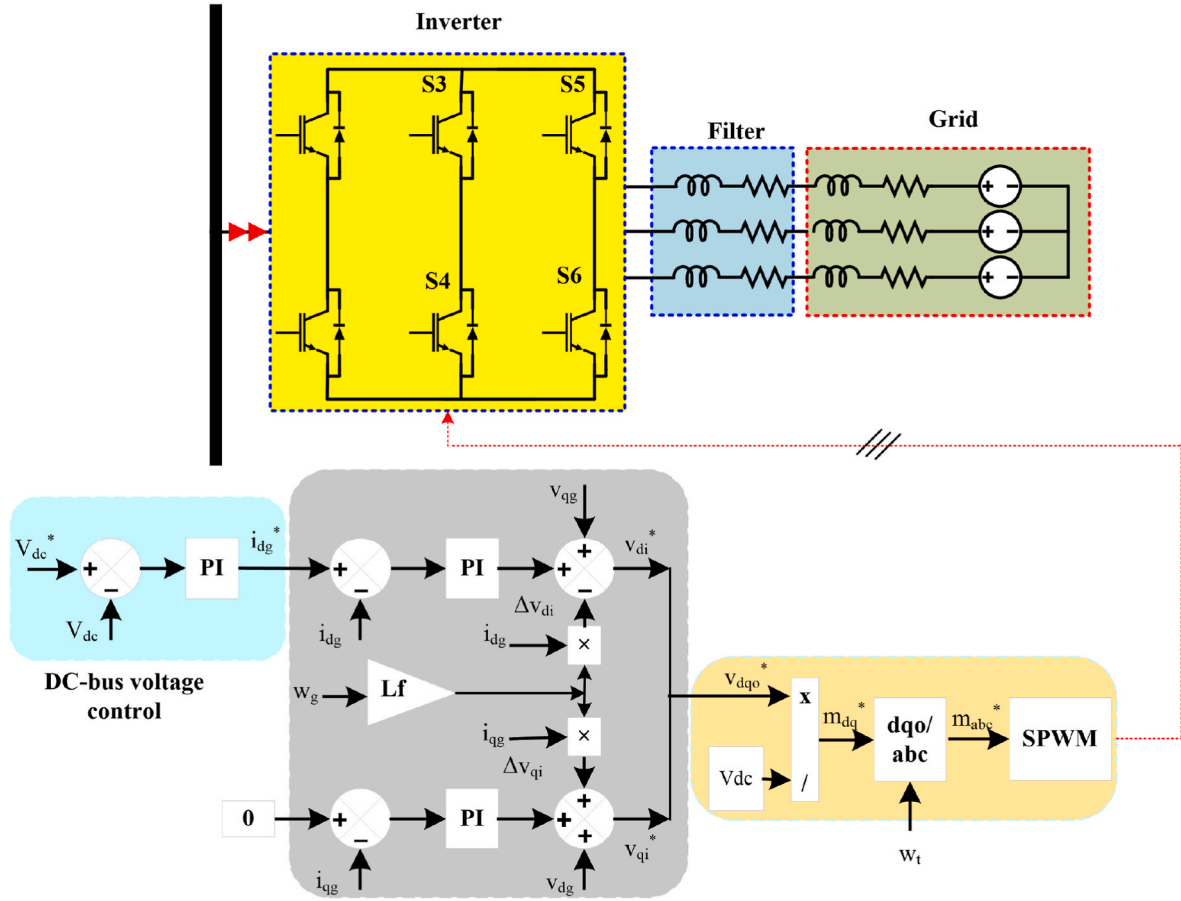


Fig. 10. Block diagram of the grid side inverter control.

inverter and the main grid. ω_g is the angular frequency of the grid. Meanwhile, v_{dg} and v_{qg} are the dq-axis of the grid voltages. It can be noticed that the filter values can be compensated by using a PI controller; hence these two previous equations are rewritten as in Eqs. (23) and (24).

$$v_{di}^* = \Delta v_{di} + v_{dg} - \omega_f \cdot L_f \cdot i_{qi} \quad (23)$$

$$v_{qi}^* = \Delta v_{qi} + v_{qg} + \omega_f \cdot L_f \cdot i_{di} \quad (24)$$

where $\Delta v_{di} = PI_{i_{di}}(\dot{i}_{di}^* - i_{di})$ & $\Delta v_{qi} = PI_{i_{qi}}(\dot{i}_{qi}^* - i_{qi})$.

From these relations, it is illustrated that the d- and q-axis currents are controlled by controlling the inverter d- and q-axis voltages v_{di} and v_{qi} . After the terms' compensation, as mentioned before, the reference value of the d-axis current is obtained from the DC-bus voltage control loop. Meanwhile, the q-axis reference current is set to zero to achieve a unity power factor. The control diagram of the grid side inverter based on the SPWM technique is shown in Fig. 10.

4. Case Study Description

The proposed design and control approaches are examined and

tested for a real case study located geographically at Ain Sokhna, Suez Governorate, Egypt (29° 30' 12" N, 32° 18' 37" E), as shown in Fig. 11. The area hosts an industrial development zone called "Sokhna industrial zone", with an area of 210 km², a part of the Suez Canal Economic Zone. It represents a major industrial and logistical hub at the southern entrance to the Suez Canal, supporting the light, medium, and heavy manufacturing, commercial business, and residential complexes.

With an annual average solar radiation and wind speed of 5.7 kWh/m²/day and 5.45 m/s, respectively, the studied area benefits from a high promise of solar and wind resources. Fig. 12 illustrates the heat map profiles of solar and wind resources over a year. A typical and representative electrical load profile of a residential community sector and industrial sector are implemented in this study to validate the proposed method. The average daily consumption of the aggregated load equals 40,500 kWh (16,500 community + 24,000 industrial) with a peak of approximately 4 MW. A day-to-day and timestep random variabilities between 10 and 20 % are considered to obtain a realistic profile of the community and industrial sectors. Fig. 13 shows the heat map of the aggregated load demand and samples of the daily load profiles, respectively.

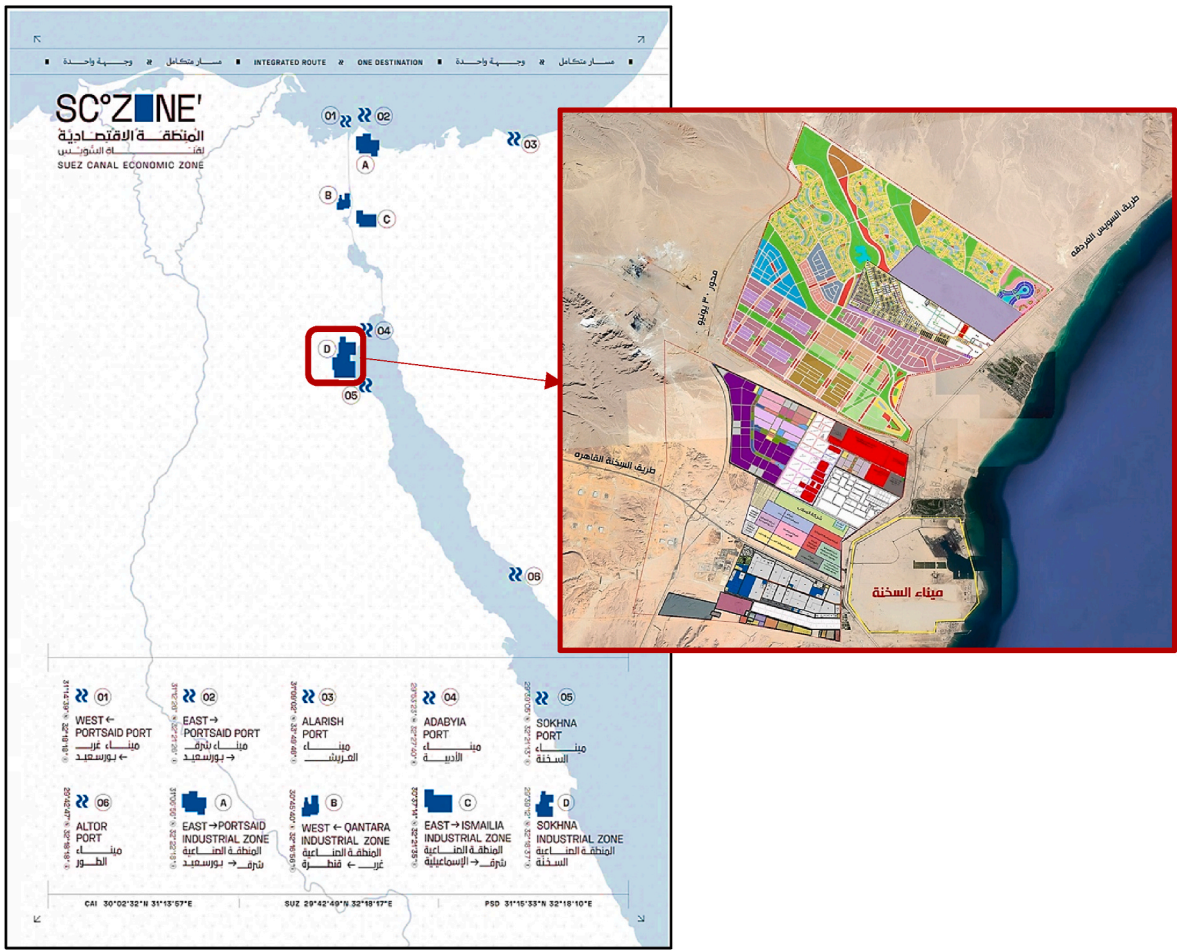


Fig. 11. Location map of the investigated area [91].

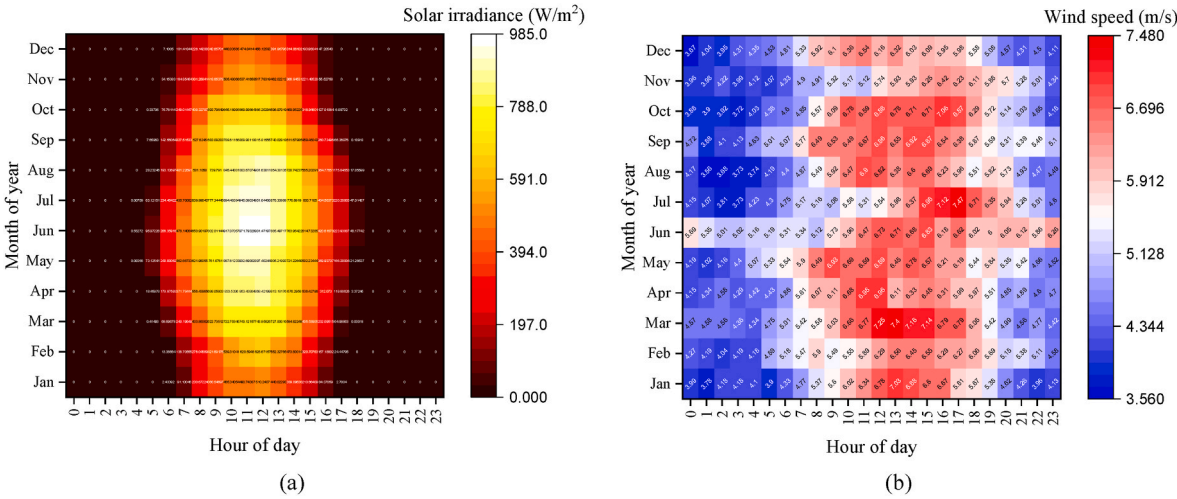


Fig. 12. Heat map of renewable resources in the case study location (a) solar irradiance and (b) wind speed.

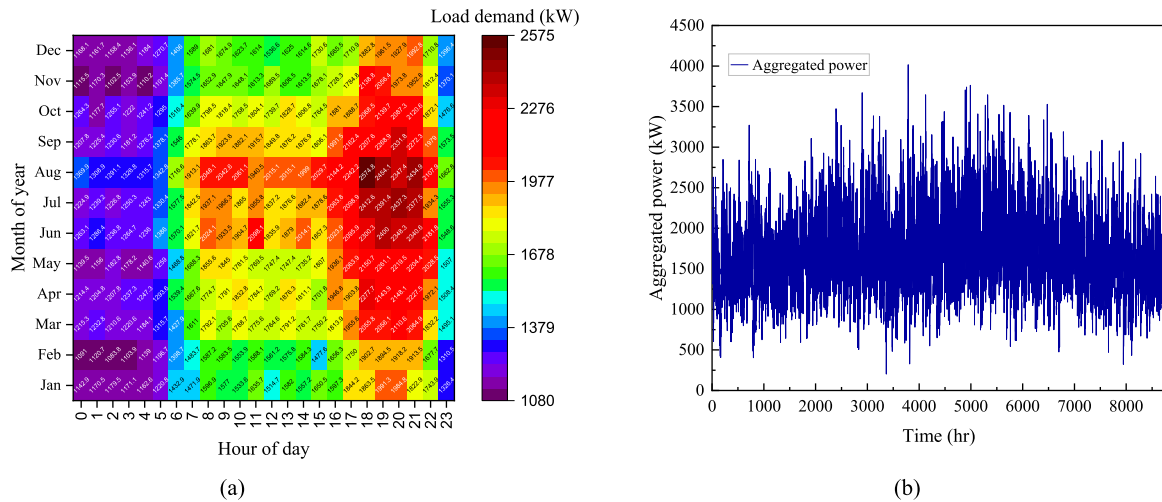


Fig. 13. Aggregated load demand of the case study (a) heat map (b) hourly profile.

5. Results and Discussion

Herein, the HRES design optimization simulation results using the HOMER Pro model and FS-MPCC/PI control results obtained from the Matlab/Simulink model are presented and analyzed in detail.

5.1. Feasibility and Design Optimization Results

According to the presented design optimization method, seven feasible power systems and the base case (grid-only system) are simulated and optimized. This includes four configurations with battery (**case#1**: solar/wind/battery/converter/grid, **case#2**: wind/battery/converter/grid, **case#3**: solar/battery/converter/grid, **case#4**: battery/converter/grid) and three configurations without battery (**case#5**: solar/wind/converter/grid, **case#6**: solar/converter/grid, **case#7**: wind/converter/grid), and a base-case (**case#8**: grid only). The optimization has been executed considering a project lifetime of 20 years, a nominal discount rate of 8.25 %, and an expected inflation rate of 4.25 % based on the market specifications on February 2022 [92]. Table 2 shows each system's optimization results, including the optimal capacity of each element and economic performance results. Moreover, Table 3 summarizes the main results concerning each feasible system's technical, energy, and environmental performance. According to the design optimization results, the main findings are summarized as follows:

- The optimal hybrid power system alternative to maintaining the demand power of the studied locality is case#5, comprising solar, wind, and converter. The system is of best economic performance

with the least TNPC of 17,371,980 \$ of and LCOE of 0.0782 \$/kWh, which is 16.6 % and 21.8 % less than the grid-only alternative of case#8.

- The winning design is a battery-less system and comprises a 2.983 MW solar park, a single 1.5 MW wind turbine, and a 2.442 MW power converter. The monthly electric production of the optimal design is given in Fig. 14. The share of solar and wind sources is found to be 31.1 and 20.5 % of the total electricity production, while the remaining (48.4 %) is purchased from the utility grid.
- From a reliability point of view, the proposed optimal design succeeds in fulfilling the 8760 h load demand, in addition to excess electricity of 358,681 kWh/day (i.e., 2.17 % of the served load) with a negligible capacity shortage of 5835 kWh/day despite the unavailability of the utility grid for five outages/year with 1.5 h/outage. Fig. 15 shows the energy scheduling of generation and consumption for three days for three different months in the year.
- Based on Table 3, the optimal design (case #5) can attain the maximum saving in carbon emission of 4,278,150 kg/year (45.8 % saving) compared to the base case. The system also has recorded an attractive social performance with JCF equals about 10 jobs, which is slightly less than case#3 of maximum JCF. Meanwhile, the winning system has recorded the second-highest value regarding the renewable fraction (RF) of 49.24 %, following case#6 (solar/converter/grid). The proposed optimal design to mitigate climate change is highly positive and supports Egypt's vision toward a sustainable and clean energy mix.
- The battery-equipped configurations of cases #1 and #2 come in the tail of alternatives rank, reflecting their nonprofit outcomes. The results reveal an almost 14 % increase in TNPC (23,763,970 \$) and

Table 2

Optimization results of the feasible alternatives indicating the optimal capacities and economic performance.

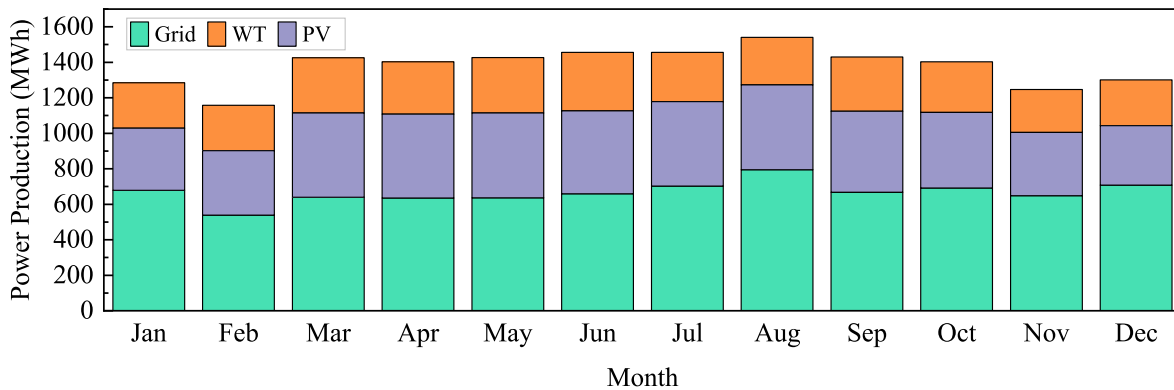
#	System configuration	Optimal capacity of system elements				Economic performance				Dispatch strategy	System rank
		PV	WT	BAT	CON	TNPC	LCOE	OPEX	CAPEX		
		kW	Qty	Qty	kW	\$	\$/kWh	\$/y	\$		
1	Solar/Wind/Battery/Converter/Grid	844	1	192	73	23,763,970	0.1141	1,469,912	3,039,331	CC	7
2	Wind/Battery/Converter/Grid	-	2	324	236	24,264,090	0.1165	1,398,891	4,540,789	CC	8
3	Solar/Battery/Converter/Grid	3782	-	12	2504	18,298,630	0.0827	1,006,634	4,105,862	LF	3
4	Battery/Converter/Grid	-	-	12	18	20,857,200	0.1001	1,477,869	0020,365	LF	6
5	Solar/Wind/Converter/Grid	2983	1	-	2442	17,371,980	0.0782	0849,194	5,398,988	CC	1
6	Solar/Converter/Grid	3774	-	-	2388	18,266,300	0.0830	1,008,399	4,048,638	CC	2
7	Wind/Converter/Grid	-	3	-	3485	18,757,240	0.0765	0823,749	7,143,000	CC	4
8	Grid only (Base Case)	-	-	-	-	20,826,160	0.1000	1,477,112	0	CC	5

Abbreviations: OpeEX: Total operational expenditure, CapEX: Total Capital Expenditure, ExEL: Excess Electricity, RenF: Renewable Fraction.

Table 3

Optimization results of the feasible alternatives indicate the technical, economic, and environmental performances.

#	System configuration	Technical performance			Generation and consumption performance				Social performance	Environmental performance
		ExEL	RenF	CShF	PV Production	WT Production	Grid Purchase	Grid Sold	JCF	Grid CO ₂ emission
		%	%	kWh/year	kWh/year	kWh/year	kWh/year	kWh/year	jobs	kg/year
1	Solar/Wind/Battery/Converter/Grid	22.55	03.46	12267.8	1,452,202	3,394,866	14,260,000	0	5.50	9,012,317
2	Wind/Battery/Converter/Grid	26.44	09.35	11181.5	-	6,789,732	13,391,690	34	4.77	8,463,550
3	Solar/Battery/Converter/Grid	0.78	38.64	7478.1	6,509,289	-	9,628,841	916,319	11.27	6,085,428
4	Battery/Converter/Grid	0.00	0.00	12704.5	-	-	14,771,120	0	1.62	9,335,350
5	Solar/Wind/Converter/Grid	2.17	49.24	5835.0	5,133,771	3,394,866	8,001,899	985,664	10.58	5,057,200
6	Solar/Converter/Grid	1.25	38.28	7860.4	6,494,879	-	9,637,845	839,967	11.25	6,091,119
7	Wind/Converter/Grid	2.91	52.74	3403.2	-	10,184,600	8,215,507	2,601,681	5.85	5,192,201
8	Grid only (Base Case)	0.00	00.00	12750.6	-	-	14,771,120	0	1.62	9,335,350

**Fig. 14.** Monthly average electric production of the optimal design

LCOE (0.1141\$/kWh) with case#1 and a 16 % increase in TNPC (24,264,090 \$) and LCOE (0.1165 \$/kWh) with case#2 compared to the base case. This is mainly attributed to the large size and, thus, the high cost of battery investment.

- Moreover, cases#1 and #2 show a low contribution from renewables in the total energy generation using low renewable fraction values of only 3.46 and 9.35 %, respectively. This is causing significant carbon emissions of 9,012,317 and 8,463,550 kg/year, respectively. For comparative purposes, these emission values are equivalent to 43.9 % (in case#1) and 40.2 % (in case#2) more than the proposed optimal design.
- Despite using a single power source with the grid, such as in case#6 (solar/converter/grid) and case#7 (wind/converter/grid), which is practically more accessible in installation and management, they are realized as not optimally economic to fulfill the load demand. The former case comes in the 2nd rank, while the latter comes in the 4th rank with TNPC values of 18,266,300 \$ and 18,757,240 \$, respectively. However, case#7 attained the best value of RF of 52.7 %, owing to the high share of wind turbine farms by 55.4 % in the total electricity production against 44.6 % purchased from the utility grid.
- From Table 3, case #3 and case # 5 have the best value for JCF with approximately 11 permanent jobs. This is mainly attributed to the large capacity of PV (~3.7 GW) which enjoy a higher employment factor. Despite the interesting social benefits compared to the other alternatives, they have lower TNPC and LCOE values than case# 5

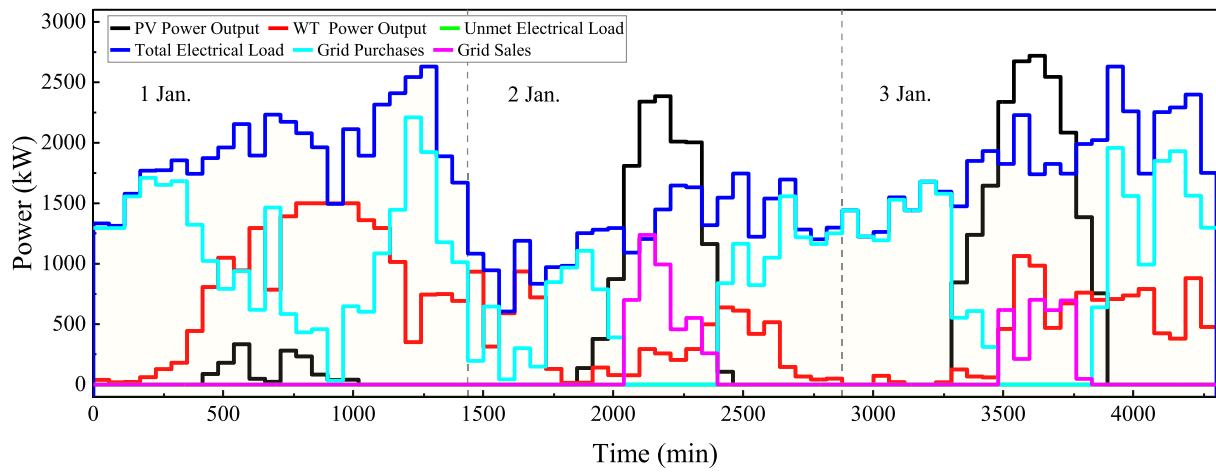
and are therefore ranked 2 and 3 in the cost analysis. On the other hand, case #4 and case 6 have recorded poor social benefits due to the absence of either solar or wind power.

5.2. Model Predictive Control Results

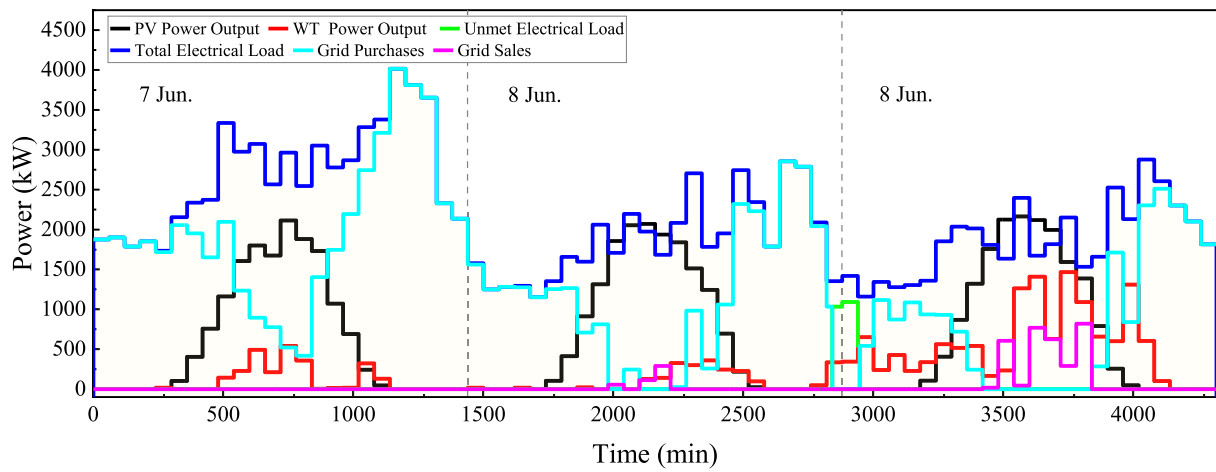
This section compares the optimal system technical performance under conventional PI and the proposed FS-MPCC techniques. The system behavior was examined under two different cases, as given below:

5.2.1. Case-1: Variable irradiance and wind velocity

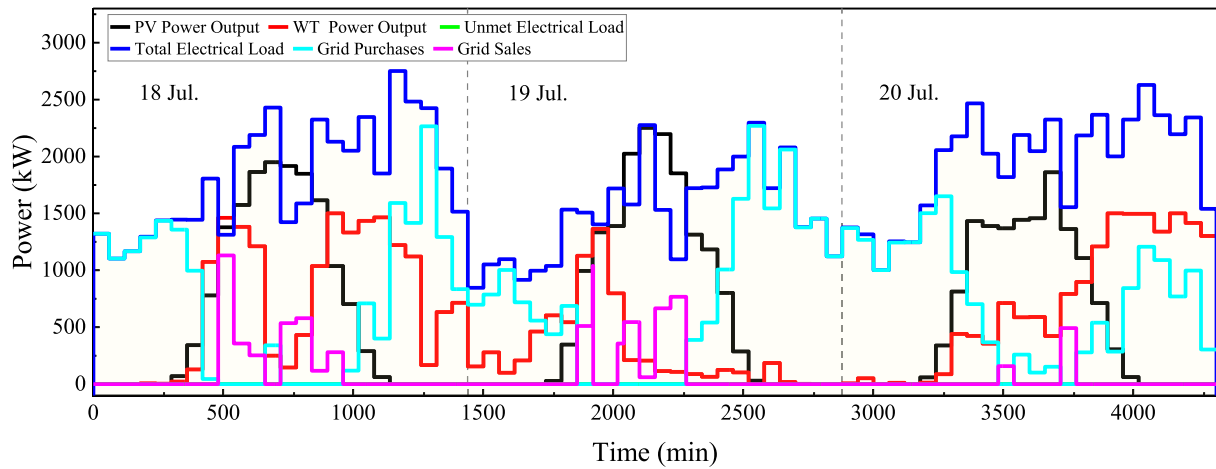
In this test, the variance in both solar irradiance and wind velocity were considered as provided in Fig. 16-a. These ever changing nature is attributed to passing clouds or sudden shadows and instant wind variation respectively. Despite, the utilized MPPT of both the PV and WT systems efficiently extracted the maximum possible energy from both solar and wind energies. The corresponding behavior of the MPPT of the PV system is shown in Fig. 16-b, from which the response of PV output power using both MPC and PI controllers is almost the same and were able to fastly responded to the reduction of solar radiation. Regarding the response of wind power coefficient it was depicted in Fig. 16-c, from which the starting response of wind power coefficient is better utilizing the MPC compared to the PI controller. In addition, during the moments of variations, both controllers' profitability responded to the increase of wind speed. Thanks to the fast response of the utilized



(a)



(b)



(c)

Fig. 15. Fifteen minutes of energy scheduling during three consecutive days in (a) January, (b) June, and (c) July.

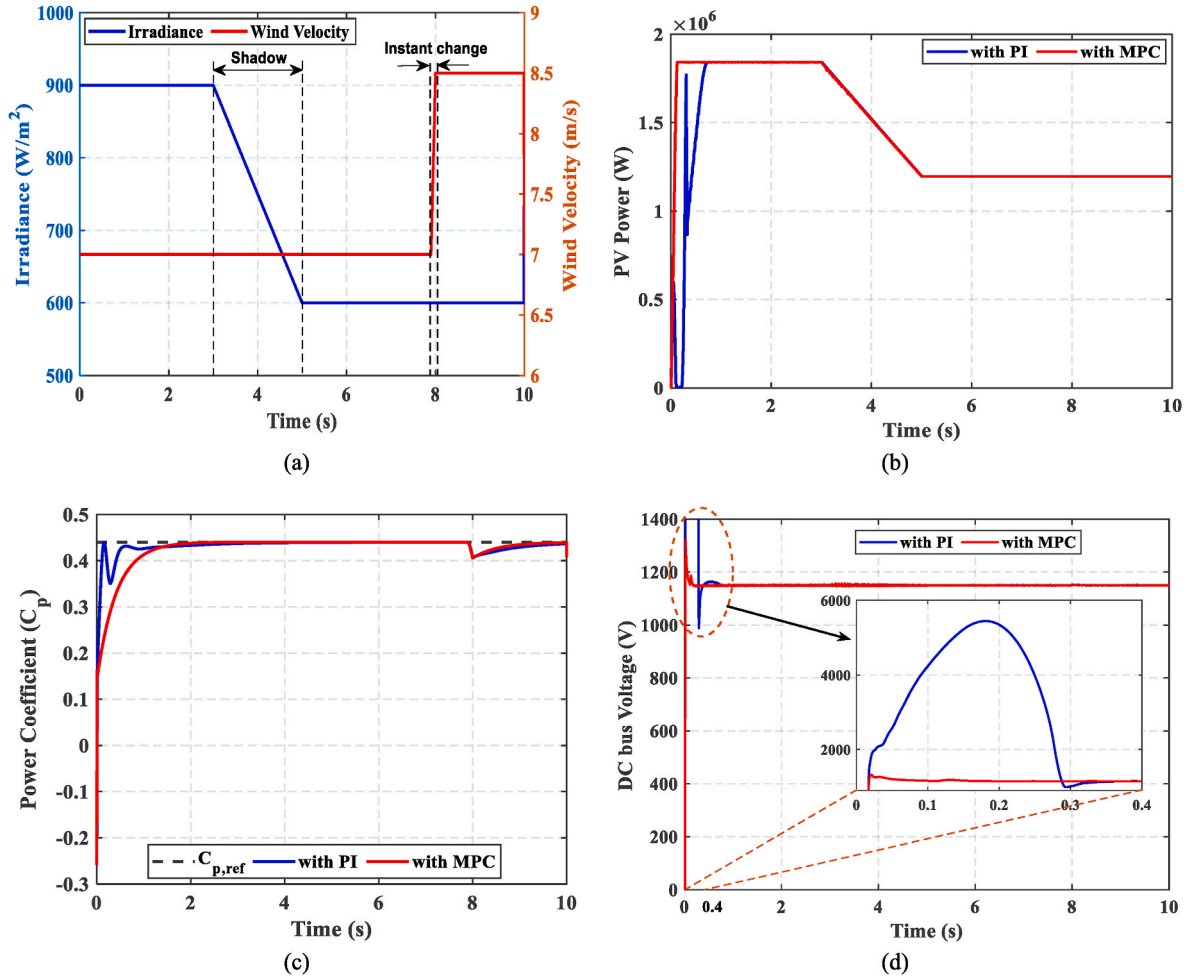


Fig. 16. Results of PI and FS-MPCC with variable irradiance and wind velocity (a) solar radiation and wind velocity profiles, (b) PV generated power, (c) wind power coefficient, and (d) DC-bus voltage behavior.

controller.

In addition, the DC-bus voltage behavior shown in Fig. 16-d demonstrates how the optimal system responded during the examined variation. It can be recognized that the dynamic/transient behavior of DC-bus voltage was significantly enhanced in terms of over- and undershoot and the steady-state response using the proposed FS-MPCC against the PI controller.

Fig. 17 indicates the resultant grid current using both controllers in which Fig. 17-a highlights the q-component and Fig. 17-b shows the d-component. From the grid side converter current, it can be seen that both controllers successfully kept the q-component at zero, the reason behind improving the system's power factor near unity. Similarly, the machine-side converter produced the current profiles portrayed in the same figure. From Fig. 17-c, it can be recognized that the I_q component profitably tracked the generated reference using both controllers with superior performance to the FS-MPCC, which significantly enhances the transient behavior of the current. Besides, in Fig. 17-d, the I_d component was successfully maintained at zero to decrease the machine losses, increase efficiency, and increase maximum torque per ampere.

Following the voltages, Fig. 18-a and Fig. 18-b signifies the resultant voltage of the machine side converter utilizing both controllers from which the efficient performance of the FS-MPCC compared with the PI controller can be recognized. Besides, to clarify the resultant power factor using both control techniques, Fig. 18-c and Fig. 18-d demonstrate the produced voltage and current of the primary grid from which it can be recognized that both voltage and current are in-phase which clarifies the effectiveness of the proposed control technique.

5.2.2. Case-2: Constant irradiance and variable wind velocity

In this examination, the solar irradiance was kept unchanged while a substantial and quick variation in the wind velocity is considered, as provided in Fig. 19-a. The MPPT of both solar and wind systems profitably obtained the maximum possible energy using the MPC, as verified in Fig. 19-b and Fig. 19-c. On the other hand, the MPPT-based-PI controller could not track the substantial and quick variation in the wind velocity as recognized in the power coefficient profile shown in Fig. 18-c; this, in turn, influences the operation of the MPPT of the PV system. During the moments of variations, the FS-MPCC profitability responded to both the reduction and increase of wind velocity thanks to its swift dynamic response. The DC-bus voltage behavior displayed in Fig. 19-d demonstrates how the employed FS-MPCC for the optimal system responded during the assessed deviation. From the DC bus voltage profiles, it can be identified that the behavior of DC-bus voltage was significantly kept steady using the proposed MPC against the PI controller, which failed to override the transient overshoot during starting and at the moment of disturbances. In addition, Fig. 19-e and Fig. 19-f signifies the resultant grid current using PI and FS-MPCC, from which Fig. 19-e emphasizes the q-axis component and Fig. 19-f reveals the d-axis component. Fig. 19-e shows that the FS-MPCC successfully kept the q-axis component at zero during the examined disturbance compared to the PI controller. Moreover, from Fig. 19-f, the d-axis component of the grid side converter was kept steady during the deviation compared to the PI controller, which suffered from a substantial deviation from the desired current. Lastly, Fig. 19-g and Fig. 19-h indicates the subsequent voltage of the machine side converter employing

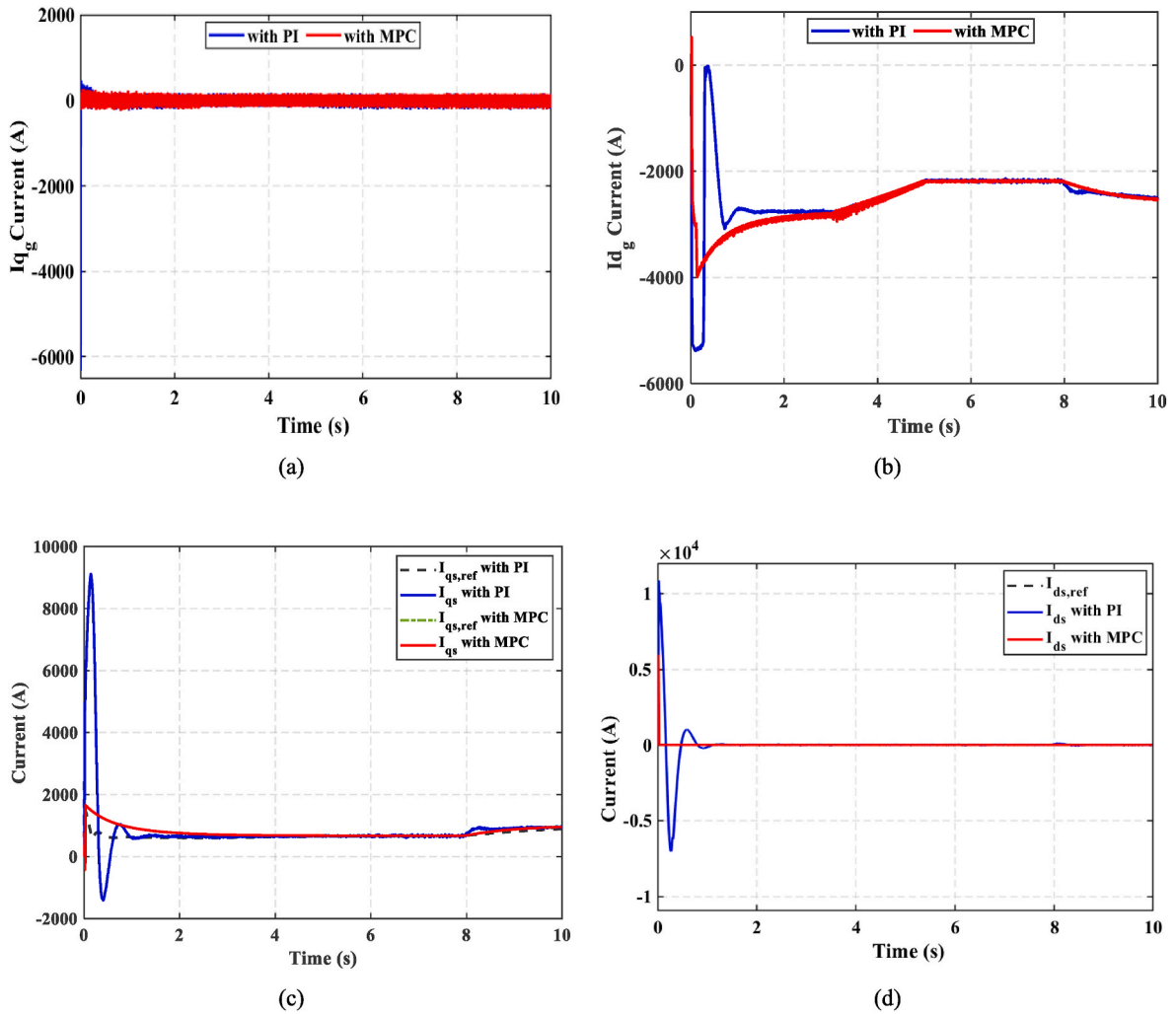


Fig. 17. (a) q-axis component of grid side converter current, (b) d-axis component of grid side converter current, (c) q-axis component of machine side converter current, (d) d-axis component of machine side converter current

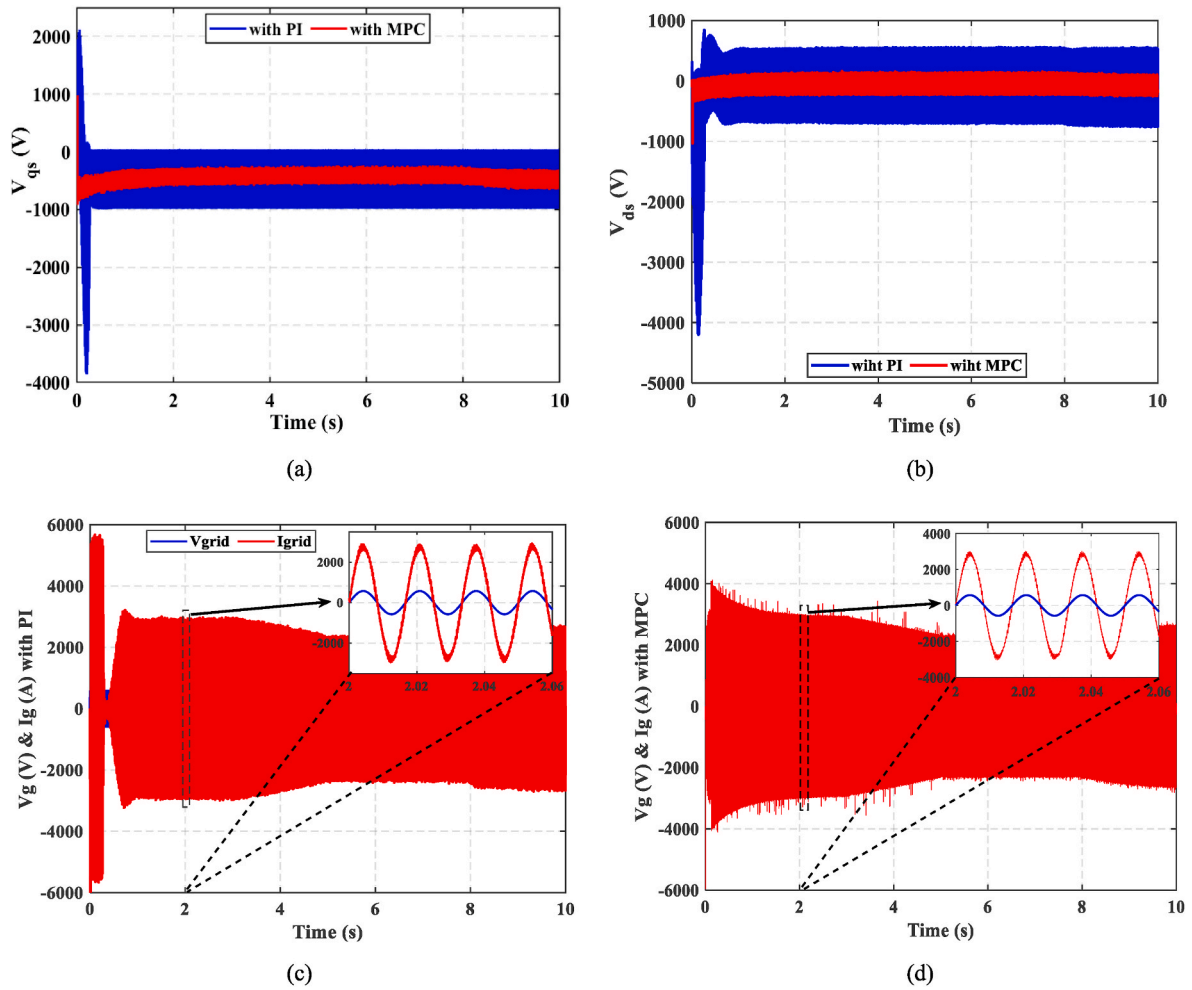


Fig. 18. (a) q-axis component of machine side converter voltage, (b) d-axis component of machine side converter voltage, (c) Grid voltages and currents using PI, and (d) Grid voltages and currents using MPC

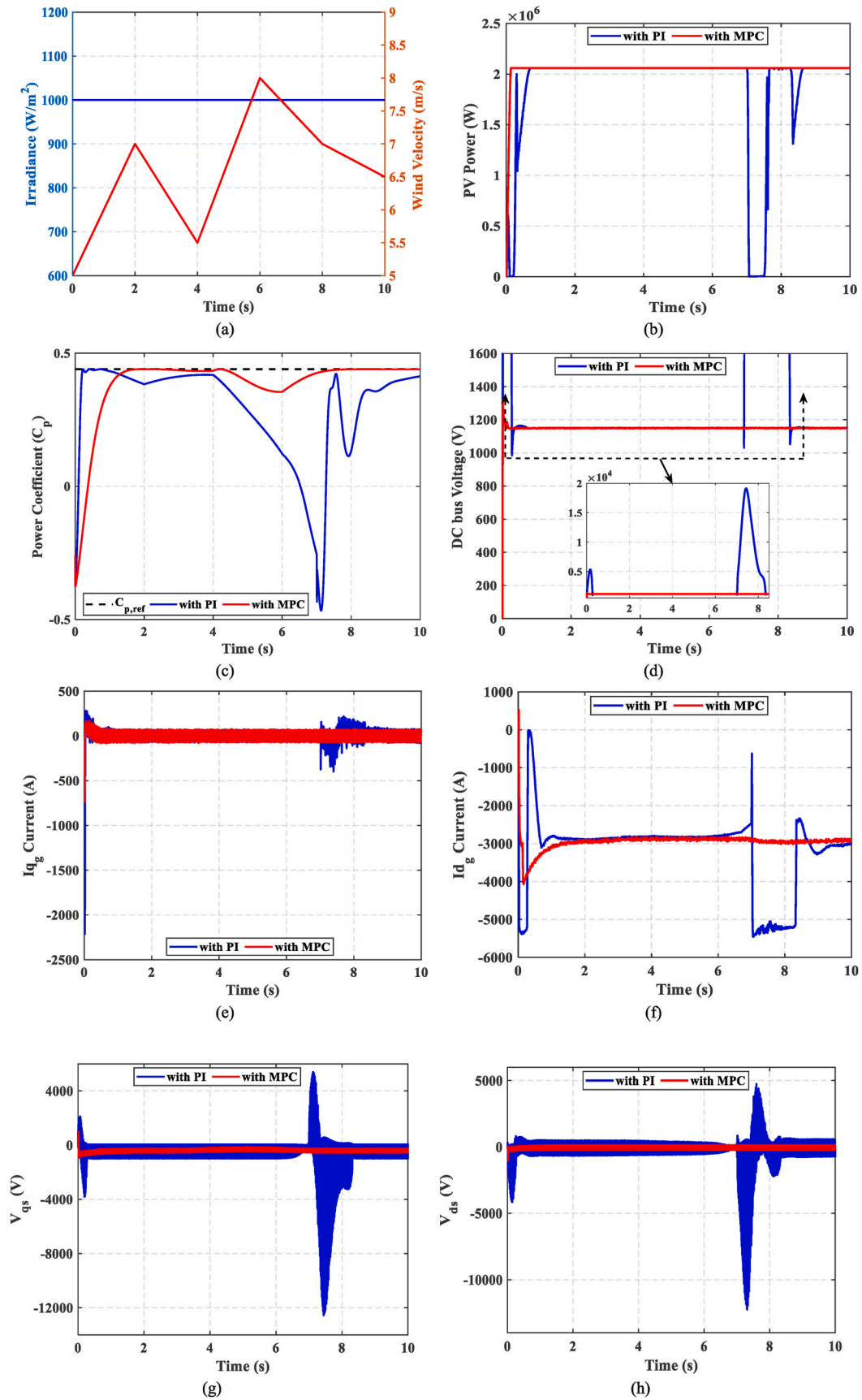


Fig. 19. Results of PI and MPC with variable wind velocity and constant irradiation (a) solar radiation and wind velocity profiles, (b) PV generated power, (c) wind power coefficient, (d) DC-bus voltage behavior, (e) q-axis component of grid side converter current, (f) d-axis component of grid side converter current, (g) q-axis component of machine side converter voltage, and (h) d-axis component of machine side converter voltage

both PI and FS-MPCC from which the competent implementation of the FS-MPCC can be distinguished over the PI controller.

6. Conclusions

This study presented a two-stage research method for techno-enviro-socio-economic design optimization and model predictive control of large-scale-grid connected hybrid solar/wind energy systems. The hybrid system is supposed to provide the electricity demand of residential and industrial loads located in Sokhna Industrial Zone, Suez city, Egypt. The first stage to obtain the optimal system design is performed using the HOMER tool. Meanwhile, the second stage of the system control is developed by Matlab/Simulink to test the finite-set MPC. In addition, the hybrid system provided power continuity and helped stabilize the overall system's dynamic response in a variety of situations where the DC bus was kept steady and constant. Moreover, the FS-MPC is adopted for the maximum power extraction from both the wind generating system and the PV system to increase the amount of the generated power and hence the efficiency. From the design point of view, the hybrid system with case# 5 that comprises a 2.983 MW solar park, a single 1.5 MW wind turbine, and a 2.442 MW power converter is of best cost, energy, emission, and social indices with the least TNPC (17,371,980 \$) of an LCOE (0.0782 \$/kWh), negligible capacity shortage (5835 kWh/day), maximum saving in CO₂ (4,278,150 kg/year), attractive job opportunities (10 jobs). This proves that the use of PV, with a percentage share of 31.1 % in the total demand, and WT, with a percentage share of 20.5 %, engenders a complementarity behavior reducing their intermittent nature that would ameliorate the system's reliability, operability, and profitability.

Concerning the system control, the proposed FS-MPC eliminated the use of the weighting factor where the current control algorithm is used. The obtained results from the FS-MPCC have been compared with the counterparts from the conventional PI controller. It was revealed that better stability and quality for the overall system performance are achieved with the FS-MPC. Also, the tracking performance for all the DC-bus voltage and the maximum power from the PV and the WT systems are much better during different climatic variation circumstances. Further, the injected current to the grid has higher quality when FS-MPC is applied rather than the PI controller. Moreover, the developed torque ripples are decreased which lead to increase the lifetime of the WT system, and the injected waveforms of the current become smooth as the DC-bus voltage is stable.

The future directions of the current study are directed to reduce the computational time and improve the FS-MPC for the WT generation system. Besides, adopting an advanced controller for the DC-bus voltage control for further stabilization and robustness. Experimental validation of the control method using a laboratory setup is also important. Faced with a growing energy crisis, greenhouse gas emissions, and increasing extreme weather events, this study could serve as a benchmark model for researchers, stakeholders, and investors who are involved in sustainable energy planning and aims to adapt to the inevitable impacts of climate change in most location world and particularly in Egypt.

CRediT authorship contribution statement

Mohamed R. Elkadeem: Conceptualization, Visualization, Methodology, Software, Formal analysis, Writing - original draft, Writing - review & editing. **Kotb M. Kotb:** Conceptualization, Methodology, Investigation, Validation, Formal analysis, Writing - original draft, Writing - review & editing. **Mohamed A. Abido:** Investigation, Writing - review & editing, Supervision. **Hany M. Hasanien:** Validation, Formal analysis, Writing - review & editing. **Eman G. Atiya:** Visualization, Formal analysis, Writing - review & editing. **Dhafer Almkhles:** Validation, Writing - review & editing. **Mahmoud F. Elmorshedy:** Methodology, Validation, Investigation, Formal analysis, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

Prof. Abido and Dr. Elkadeem would like to acknowledge the support provided by the Interdisciplinary Research Center of Renewable Energy and Power Systems (IRC-REPS), KFUPM. Also, Dr. Almkhles and Dr. Elmorshedy acknowledge the support of Prince Sultan University during this research.

Appendix: See Table A1 for technical parameters and cost data of the system's components

Table A1
Technical parameters and cost data of the system's components

Element/parameter	Value	Element/parameter	Value
Wind turbine (Nordex S70 1500 kW)		Power converter (Generic)	
Rated capacity	1500 kW	Inversion efficiency	95 %
Cut-in wind speed	4 m/s	Rectification efficiency	95 %
Cut-out wind speed	20 m/s	Rectifier relative capacity	100 %
Rated wind speed	25 m/s	Capital cost	\$300/kW
Hub height	30 m	Replacement cost	\$300/kW
Rotor diameter	70 m	O&M cost	\$0/yr
Capital cost	\$2,032,500	Lifetime	15 years
Replacement cost	\$1,829,250	Battery storage (Surrette 4 KS 25P)	
O&M cost	\$20,325/year	Nominal voltage	4 V
Lifetime	20 years	Nominal capacity	7.55 kWh
Photovoltaic module (Trina Solar310TSM-310PDG14)		Capacity	1350 Ah
Rated capacity	0.31 kW	Rate constant	0.254
Temperature coefficient	−0.4882/°C	Maximum capacity	1890 Ah
Operating temperature	46.7 °C	SOC limits	40–100 %
Efficiency at STC	13 %	Roundtrip efficiency	80 %
Derating factor	85 %	Maximum charge current	459 A
		Maximum discharge current	459 A

(continued on next page)

Table A1 (continued)

Element/parameter	Value	Element/parameter	Value
Wind turbine (Nordex S70 1500 kW)		Power converter (Generic)	
Nameplate voltage	45.5 V	Capital cost	\$1250/kW
Nameplate current	8.85 A	Replacement cost	\$1100/kW
Ground reflection	0 %	O&M cost	\$ 0/yr
Derating factor	85 %	Lifetime	10 years
Capital cost	\$883/kW	Throughput	10,551.7
Replacement cost	\$883/kW	String size	12
O&M cost	\$15/year	Utility grid	
Lifetime	20 years	Purchase rate	\$0.1/kWh
		Sellback rate	\$0.03/kWh
		Mean outage frequency	5/year
		Mean repair time	1.5 h
		Repair time variability	30 %

References

- Shafiullah M, Ahmed SD, Al-Sulaiman FA. Grid integration challenges and solution strategies for solar PV systems: a review. *IEEE Access* 2022;10:52233–57. <https://doi.org/10.1109/access.2022.3174555>.
- The World Bank. Access to electricity (% of population). 2022. 2022.
- International Energy Agency. Electricity Market Report. 2022. January 2022, <https://www.iea.org/reports/electricity-market-report-january-2022>.
- International Renewable Energy Agency (IRENA). World energy transitions outlook 2022. 2022. <https://www.irena.org/publications/2022/Mar/World-Energy-Transitions-Outlook-2022>.
- Kotb KM, Elmorshedy MF, Salama HS, Dán A. Enriching the stability of solar/wind DC microgrids using battery and superconducting magnetic energy storage based fuzzy logic control. *J Energy Storage* 2022;45. <https://doi.org/10.1016/j.est.2021.103751>.
- Guo Z, Wei W, Chen L, Dong ZY, Mei S. Impact of energy storage on renewable energy Utilization: a geometric description. *IEEE Trans Sustain Energy* 2021;12: 874–85. <https://doi.org/10.1109/TSTE.2020.3023498>.
- Hu J, Harmsen R, Crijns-Graus W, Worrell E. Barriers to investment in utility-scale variable renewable electricity (VRE) generation projects. *Renew Energy* 2018;121: 730–44. <https://doi.org/10.1016/j.renene.2018.01.092>.
- Elkadeem MR, Younes A, Sharshir SW, Campana PE, Wang S. Sustainable siting and design optimization of hybrid renewable energy system: a geospatial multi-criteria analysis. *Appl Energy* 2021;295:117071. <https://doi.org/10.1016/j.apenergy.2021.117071>.
- Mazzeo D, Matera N, De Luca P, Baglivo C, Congedo PM, Oliveti G. A literature review and statistical analysis of photovoltaic-wind hybrid renewable system research by considering the most relevant 550 articles: an upgradable matrix literature database. *J Clean Prod* 2021;295:126070. <https://doi.org/10.1016/j.jclepro.2021.126070>.
- Thirunavukkarasu GS, Seyedmahmoudian M, Jamei E, Horan B, Mekhilef S, Stojcevski A. Role of optimization techniques in microgrid energy management systems—a review. *Energy Strategy Rev* 2022;43:100899. <https://doi.org/10.1016/j.esr.2022.100899>.
- Roslan MF, Hannan MA, Ker PJ, Mannan M, Muttaqi KM, Mahlia TI. Microgrid control methods toward achieving sustainable energy management: a bibliometric analysis for future directions. *J Clean Prod* 2022;348:131340. <https://doi.org/10.1016/j.jclepro.2022.131340>.
- Kotb KM, Said SM, Dan A, Hartmann B. Stability enhancement of isolated-microgrid applying solar power generation using SMES based FLC. In: 2019 7th int. Istanbul smart grids cities Congr. Fair. IEEE; 2019. p. 104–8. <https://doi.org/10.1109/SGCF.2019.8782321>.
- Dagar A, Gupta P, Niranjana V. Microgrid protection: a comprehensive review. *Renew Sustain Energy Rev* 2021;149:111401. <https://doi.org/10.1016/j.rser.2021.111401>.
- Twaha S, Ramli MAM. A review of optimization approaches for hybrid distributed energy generation systems: off-grid and grid-connected systems. *Sustain Cities Soc* 2018;41:320–31. <https://doi.org/10.1016/j.scs.2018.05.027>.
- Ullah Z, Elkadeem MR, Kotb KM, Taha IBM, Wang S. Multi-criteria decision-making model for optimal planning of on/off grid hybrid solar, wind, hydro, biomass clean electricity supply. *Renew Energy* 2021;179:885–910. <https://doi.org/10.1016/j.renene.2021.07.063>.
- Elkadeem MR, Kotb KM, Ullah Z, Atiya EG, Dán A, Wang S. A two-stage multi-attribute analysis method for city-integrated hybrid mini-grid design. *Sustain Cities Soc* 2020;65:102603. <https://doi.org/10.1016/j.scs.2020.102603>.
- Lian J, Zhang Y, Ma C, Yang Y, Chaima E. A review on recent sizing methodologies of hybrid renewable energy systems. *Energy Convers Manag* 2019;199:112027. <https://doi.org/10.1016/j.enconman.2019.112027>.
- Thirunavukkarasu M, Sawle Y, Lala H. A comprehensive review on optimization of hybrid renewable energy systems using various optimization techniques Deficiency of Power Supply Probability. *Renew Sustain Energy Rev* 2023;176:113192. <https://doi.org/10.1016/j.rser.2023.113192>.
- Mazzeo D, Matera N, De Luca P, Baglivo C, Congedo PM, Oliveti G. A literature review and statistical analysis of photovoltaic-wind hybrid renewable system research by considering the most relevant 550 articles: an upgradable matrix literature database. *J Clean Prod* 2021;295:126070. <https://doi.org/10.1016/j.jclepro.2021.126070>.
- Riayatsyah TMI, Geumpana TA, Fattah IMR, Rizal S, Mahlia TMI. Techno-economic analysis and optimisation of campus grid-connected hybrid renewable energy system using HOMER grid. *Sustainability* 2022;14:7735. <https://doi.org/10.3390/su14137735>.
- Morad M, Nayel M, Elbaset AA, Galal AIA. Sizing and analysis of grid-connected microgrid system for assiut university using HOMER software. In: 2018 twent. Int. Middle east power syst. Conf. IEEE; 2018. p. 694–9. <https://doi.org/10.1109/MEPCON.2018.8635166>.
- Zhang X, Wei QS, Oh BS. Cost analysis of off-grid renewable hybrid power generation system on Ui Island, South Korea. *Int J Hydrogen Energy* 2022;47: 13199–212. <https://doi.org/10.1016/j.ijhydene.2022.01.150>.
- Alsagri AS, Alrobaian AA, Nejlaoui M. Techno-economic evaluation of an off-grid health clinic considering the current and future energy challenges: a rural case study. *Renew Energy* 2021;169:34–52. <https://doi.org/10.1016/j.renene.2021.01.017>.
- Vichos E, Sifakis N, Tsoutsos T. Challenges of integrating hydrogen energy storage systems into nearly zero-energy ports. *Energy* 2022;241:122878. <https://doi.org/10.1016/j.ENERGY.2021.122878>.
- Al-Najjar H, Pfeifer C, Al Afif R, El-Khozondar HJ. Performance evaluation of a hybrid grid-connected photovoltaic biogas-generator power system. *Energies* 2022; 15:3151. <https://doi.org/10.3390/en15093151>.
- Khalil L, Liaquat Bhatti K, Arslan Iqbal Awan M, Riaz M, Khalil K, Alwaz N. Optimization and designing of hybrid power system using HOMER pro. *Mater Today Proc* 2021;47:S110–5. <https://doi.org/10.1016/j.matpr.2020.06.054>.
- Azerefeg TM, Bhandari R, Ramayya AV. Techno-economic analysis of grid-integrated PV/wind systems for electricity reliability enhancement in Ethiopian industrial park. *Sustain Cities Soc* 2020;53:101915. <https://doi.org/10.1016/j.scs.2019.101915>.
- Dahiru AT, Tan CW. Optimal sizing and techno-economic analysis of grid-connected nanogrid for tropical climates of the Savannah. *Sustain Cities Soc* 2020; 52:101824. <https://doi.org/10.1016/j.scs.2019.101824>.
- Bhimaraju A, Mahesh A, Joshi SN. Techno-economic optimization of grid-connected solar-wind-pumped storage hybrid energy system using improved search space reduction algorithm. *J Energy Storage* 2022;52:104778. <https://doi.org/10.1016/j.est.2022.104778>.
- Sharma A, Agrawal S, Urpelainen J. The adoption and use of solar mini-grids in grid-electrified Indian villages. *Energy Sustain Dev* 2020;55:139–50. <https://doi.org/10.1016/j.esd.2020.01.005>.
- Suresh V, Muralidhar M, Kiranmayi R. Modelling and optimization of an off-grid hybrid renewable energy system for electrification in a rural areas. *Energy Rep* 2020;6:594–604. <https://doi.org/10.1016/j.egyr.2020.01.013>.
- Hadidian Moghaddam MJ, Kalam A, Nowdeh SA, Ahmadi A, Babanezhad M, Saha S, et al. Optimal sizing and energy management of stand-alone hybrid photovoltaic/wind system based on hydrogen storage considering LOEE and LOLE reliability indices using flower pollination algorithm. *Renew Energy* 2019;135: 1412–34. <https://doi.org/10.1016/j.renene.2018.09.078>.
- Jahangiri M, Rizi RA, Shamsabadi AA. Feasibility study on simultaneous generation of electricity and heat using renewable energies in Zarrin Shahr, Iran. *Sustain Cities Soc* 2018;38:647–61. <https://doi.org/10.1016/j.scs.2018.01.043>.
- Delano O, Odou T, Bhandari R, Adamou R. Hybrid off-grid renewable power system for sustainable rural electrification in Benin. *Renew Energy* 2020;145: 1266–79. <https://doi.org/10.1016/j.renene.2019.06.032>.
- Liu J, Wang M, Peng J, Chen X, Cao S, Yang H. Techno-economic design optimization of hybrid renewable energy applications for high-rise residential buildings. *Energy Convers Manag* 2020;213:112868. <https://doi.org/10.1016/j.enconman.2020.112868>.

- [36] Ali F, Ahmar M, Jiang Y, AlAhmad M. A techno-economic assessment of hybrid energy systems in rural Pakistan. *Energy* 2021;215:119103. <https://doi.org/10.1016/j.energy.2020.119103>.
- [37] Kamran M, Asghar R, Mudassar M, Abid MI. Designing and economic aspects of run-of-canal based micro-hydro system on Balloki-Sulaimanki Link Canal-I for remote villages in Punjab, Pakistan. *Renew Energy* 2019;141:76–87. <https://doi.org/10.1016/j.renene.2019.03.126>.
- [38] Li Q, Loy-Benitez J, Nam KJ, Hwangbo S, Rashidi J, Yoo CK. Sustainable and reliable design of reverse osmosis desalination with hybrid renewable energy systems through supply chain forecasting using recurrent neural networks. *Energy* 2019;178:277–92. <https://doi.org/10.1016/j.energy.2019.04.114>.
- [39] Abdelshafy AM, Jurasz J, Hassan H, Mohamed AM. Optimized energy management strategy for grid connected double storage (pumped storage-battery) system powered by renewable energy resources. *Energy* 2020;192:116615. <https://doi.org/10.1016/j.energy.2019.116615>.
- [40] Elmorshedy MF, Elkadeem MR, Kotb KM, Taha IBM, El-Nemr MK, Kandeal AW, et al. Feasibility study and performance analysis of microgrid with 100% hybrid renewables for a real gricultural irrigation application. *Sustain Energy Technol Assessments* 2022;53:102746. <https://doi.org/10.1016/j.seta.2022.102746>.
- [41] Tiwary A, Spasova S, Williams ID. A community-scale hybrid energy system integrating biomass for localised solid waste and renewable energy solution: evaluations in UK and Bulgaria. *Renew Energy* 2019;139:960–7. <https://doi.org/10.1016/j.renene.2019.02.129>.
- [42] Luta DN, Raji AK. Decision-making between a grid extension and a rural renewable off-grid system with hydrogen generation. *Int J Hydrogen Energy* 2018;43: 9535–48. <https://doi.org/10.1016/j.ijhydene.2018.04.032>.
- [43] Mavromatidis G, Orehounig K, Carmeliet J. Comparison of alternative decision-making criteria in a two-stage stochastic program for the design of distributed energy systems under uncertainty. *Energy* 2018;156:709–24. <https://doi.org/10.1016/j.energy.2018.05.081>.
- [44] Chaichan W, Waewsak J, Nikhom R, Kongruang C, Chiwamongkhonkarn S, Gagnon Y. Optimization of stand-alone and grid-connected hybrid solar/wind/fuel cell power generation for green islands: application to Koh Samui, southern Thailand. *Energy Rep* 2022;8:480–93. <https://doi.org/10.1016/j.egyr.2022.07.024>.
- [45] Ghathani AM, Al-Hanbali A, Mohammed A, Attia AM, Saleh H, Alsawafy O. Optimization of a solar-wind-grid powered desalination system in Saudi Arabia. *Renew Energy* 2021;178:295–306. <https://doi.org/10.1016/j.renene.2021.06.060>.
- [46] Sanni SO, Oricha JY, Oyewole TO, Bawonda FI. Analysis of backup power supply for unreliable grid using hybrid solar PV/diesel/biogas system. *Energy* 2021;227: 120506. <https://doi.org/10.1016/j.energy.2021.120506>.
- [47] Elmorshedy MF, Xu W, El-Sousy FFM, Islam MR, Ahmed AA. Recent achievements in model predictive control techniques for industrial motor: a comprehensive state-of-the-art. *IEEE Access* 2021;9:58170–91. <https://doi.org/10.1109/ACCESS.2021.3073020>.
- [48] Jlassi I, Cardoso AJM. Fault-tolerant back-to-back converter for direct-drive PMSG wind turbines using direct torque and power control techniques. *IEEE Trans Power Electron* 2019;34:11215–27. <https://doi.org/10.1109/TPEL.2019.2897541>.
- [49] Zarei ME, Ramirez D, Prodanovic M, Arana GM. Model predictive control for PMSG-based wind turbines with overmodulation and adjustable dynamic response time. *IEEE Trans Ind Electron* 2022;69:1573–85. <https://doi.org/10.1109/TIE.2021.3057021>.
- [50] Elmorshedy MF, Xu W, El-Sousy FFM, Islam MR, Ahmed AA. Recent achievements in model predictive control techniques for industrial motor: a comprehensive state-of-the-art. *IEEE Access* 2021;9:58170–91. <https://doi.org/10.1109/ACCESS.2021.3073020>.
- [51] Rodriguez J, Fellow L, Garcia C, Mora A, Davari SA, Member S, et al. Latest advances of model predictive control in electrical drives — Part II : applications and benchmarking with classical control. *Methods* 2022;37:5047–61.
- [52] Harbi I, Ahmed M, Heldwein ML, Kennel R, Abdelrahman M. Enhanced Fault-tolerant robust deadbeat predictive control for nine-level ANPC-based converter. *IEEE Access* 2022;10:108492–505. <https://doi.org/10.1109/ACCESS.2022.3213322>.
- [53] Moon HC, Lee JS, Lee KB. A robust deadbeat finite set model predictive current control based on discrete space vector modulation for a grid-connected voltage source inverter. *IEEE Trans Energy Convers* 2018;33:1719–28. <https://doi.org/10.1109/TEC.2018.2830776>.
- [54] Zhou J, Hassan MA, Zhang J, Zhang Z, Wu S, Hou M, et al. A novel continuous control set model predictive control to guarantee stability and robustness for buck power converter in DC microgrids. *Energy Rep* 2021;7:1400–15. <https://doi.org/10.1016/j.egyr.2021.09.101>.
- [55] Khan Prince MK, Arif MT, Gargoom A, Altaf MW, Haque ME. Direct model predictive control of grid connected PMSG based wind energy system with LCL filter. *Conf Rec - IAS Annu Meet (IEEE Int Appl Soc)* 2022:2022. <https://doi.org/10.1109/IAS54023.2022.9940101>. Octob:1–8.
- [56] Habib HUR, Wang S, Elkadeem MR, Elmorshedy MF. Design optimization and model predictive control of a standalone hybrid renewable energy system: a case study on a small residential load in Pakistan. *IEEE Access* 2019;7:117369–90. <https://doi.org/10.1109/ACCESS.2019.2936789>.
- [57] Kalamian N, Niri MF, Masoomfar M. Design of a nonlinear model-based predictive controller for a wind turbine based on PMSG using an augmented extended kalman filter. *Proc - 2019 6th Int Conf Control Instrum Autom ICCIA* 2019 2019. <https://doi.org/10.1109/ICCIA49288.2019.9030837>.
- [58] Mito MT, Ma X, Albulflasa H, Davies PA. Variable operation of a renewable energy-driven reverse osmosis system using model predictive control and variable recovery: towards large-scale implementation. *Desalination* 2022;532:115715. <https://doi.org/10.1016/j.desal.2022.115715>.
- [59] Milev K, Alshammari F, Yaramasu V, Duran MJ, Yadlapati K. Predictive control with fixed switching frequency for three-level boost and NPC converters interfaced PMSG wind turbine. 3rd int conf energy. *Power Environ Towar Clean Energy Technol ICEPE* 2020. <https://doi.org/10.1109/ICEPE50861.2021.9404476>. 2021: 5–10.
- [60] Wang Y, Zhang Z, Garcia C, Rodriguez J, Kennel R. Robust predictive control of grid-side power converters for PMSG wind turbine systems with stability analysis. *Proc IEEE Int Conf Ind Technol* 2020;2020. <https://doi.org/10.1109/ICIT45562.2020.9067307>. Febru:763–8.
- [61] Milev K, Yaramasu V, Dekka A, Kouro S. Modulated predictive current control of PMSG-based wind energy systems. 2020 11th power electron drive syst technol conf PEDSTC 2020. <https://doi.org/10.1109/PEDSTC49159.2020.9088365>; 2020.
- [62] Yang W, Kang X, Wang X, Wang M. MPC-based three-phase unbalanced power coordination control method for microgrid clusters. *Energy Rep* 2023;9:1830–41. <https://doi.org/10.1016/j.egyr.2022.12.079>.
- [63] Jiao F, Zou Y, Zhang X, Zhang B. Online optimal dispatch based on combined robust and stochastic model predictive control for a microgrid including EV charging station. *Energy* 2022;247:123220. <https://doi.org/10.1016/j.energy.2022.123220>.
- [64] Ouammi A. Model predictive control for optimal energy management of connected cluster of microgrids with net zero energy multi-greenhouses. *Energy* 2021;234: 121274. <https://doi.org/10.1016/j.energy.2021.121274>.
- [65] Lin Y, Zhang J, Li L. A model predictive control approach to a water pumping system in an Australian cotton farm microgrid. *Clean Energy Syst* 2022;3:100026. <https://doi.org/10.1016/j.cles.2022.100026>.
- [66] Alarcón MA, Alarcón RG, González AH, Ferramosca A. Economic model predictive control for energy management of a microgrid connected to the main electrical grid. *J Process Control* 2022;117:40–51. <https://doi.org/10.1016/j.jprocont.2022.07.004>.
- [67] Abdollahi Arjanaki A, Dehestani Kolagar A, Alizadeh Pahlavani MR. A two-level power management strategy in a DC-coupled hybrid microgrid powered by fuel cell and energy storage systems with model predictive controlled interface converter. *J Energy Storage* 2022;52:104861. <https://doi.org/10.1016/j.est.2022.104861>.
- [68] Cai W, Kordabad AB, Gros S. Energy management in residential microgrid using model predictive control-based reinforcement learning and Shapley value. *Eng Appl Artif Intell* 2023;119:105793. <https://doi.org/10.1016/j.engappai.2022.105793>.
- [69] Majji RK, Mishra JP, Dongre AA. Model predictive control based autonomous DC microgrid integrated with solar photovoltaic system and composite energy storage. *Sustain Energy Technol Assessments* 2022;54:102862. <https://doi.org/10.1016/j.seta.2022.102862>.
- [70] Vafaie MH, Mirzaei Dehkordi B, Moallem P, Kiyomarsi A. Minimizing torque and flux ripples and improving dynamic response of PMSM using a voltage vector with optimal parameters. *IEEE Trans Ind Electron* 2016;63:3876–88. <https://doi.org/10.1109/TIE.2015.2497251>.
- [71] Errouissi R, Al-Durra A, Mueen SM, Leng S. Continuous-time model predictive control of a permanent magnet synchronous motor drive with disturbance decoupling. *IET Electr Power Appl* 2017;11:697–706. <https://doi.org/10.1049/iet-epa.2016.0499>.
- [72] Das P, Abdelrahman M, Farhan A, Ismail MA, Kennel R. Predictive direct torque control of permanent magnet synchronous generators (PMSGs) without weighting factors. 2019 IEEE conf. Power electron. *Renew. Energy, IEEE*; 2019. p. 296–301. <https://doi.org/10.1109/CPERE45374.2019.8980112>.
- [73] Elmorshedy MF, Xu W, Allam SM, Rodriguez J, Garcia C. MTPA-based finite-set model predictive control without weighting factors for linear induction machine. *IEEE Trans Ind Electron* 2021;68:2034–47. <https://doi.org/10.1109/TIE.2020.2972432>.
- [74] Vazquez S, Rodriguez J, Rivera M, Franquelo LG, Norambuena M. Model predictive control for power converters and drives: advances and trends. *IEEE Trans Ind Electron* 2017;64:935–47. <https://doi.org/10.1109/TIE.2016.2625238>.
- [75] Yaramasu V, Dekka A, Yadlapati K. Fixed switching frequency predictive control of boost converter interfaced PMSG wind turbines. *Int Conf Power Electron IoT Appl Renew Energy Its Control PARC* 2020. <https://doi.org/10.1109/PARC49193.2020.236667>. 2020 2020:518–23.
- [76] Zhang Z, Babayomi O, Dragicevic T, Heydari R, Garcia C, Rodriguez J, et al. Advances and opportunities in the model predictive control of microgrids: Part I—primary layer. *Int J Electr Power Energy Syst* 2022;134:107411. <https://doi.org/10.1016/j.ijepes.2021.107411>.
- [77] Msigwa G, Ighalo JO, Yap P-S. Considerations on environmental, economic, and energy impacts of wind energy generation: projections towards sustainability initiatives. *Sci Total Environ* 2022;849:157755. <https://doi.org/10.1016/j.scitotenv.2022.157755>.
- [78] Ahmad T, Zhu H, Zhang D, Tariq R, Bassam A, Ullah F, et al. Energetics Systems and artificial intelligence: applications of industry 4.0. *Energy Rep* 2022;8:334–61. <https://doi.org/10.1016/j.egyr.2021.11.256>.
- [79] Liu Z, Sun Y, Xing C, Liu J, He Y, Zhou Y, et al. Artificial intelligence powered large-scale renewable integrations in multi-energy systems for carbon neutrality transition: challenges and future perspectives. *Energy AI* 2022;10:100195. <https://doi.org/10.1016/j.egyai.2022.100195>.
- [80] Kozlov I, Kovalchuk V, Klymchuk O, Prokopovych L. Estimates of environmental impact on the environment during energy generation. *Lect. Notes Civ. Eng.* 2023; 290:195–204. https://doi.org/10.1007/978-3-031-14141-6_19. LNCE.

- [81] Temiz M, Dincer I. Development of solar and wind based hydrogen energy systems for sustainable communities. *Energy Convers Manag* 2022;269:116090. <https://doi.org/10.1016/j.enconman.2022.116090>.
- [82] Rahman MM, Khan MMUH, Ullah MA, Zhang X, Kumar A. A hybrid renewable energy system for a North American off-grid community. *Energy* 2016;97:151–60. <https://doi.org/10.1016/j.energy.2015.12.105>.
- [83] Lyden A, Pepper R, Tuohy PG. A modelling tool selection process for planning of community scale energy systems including storage and demand side management. *Sustain Cities Soc* 2018;39:674–88. <https://doi.org/10.1016/j.scs.2018.02.003>.
- [84] Ramesh M, Saini RP. Dispatch strategies based performance analysis of a hybrid renewable energy system for a remote rural area in India. *J Clean Prod* 2020;259:120697. <https://doi.org/10.1016/j.jclepro.2020.120697>.
- [85] Elmorshedy MF, Elkadeem MR, Kotb KM, Taha IBM, Mazzeo D. Optimal design and energy management of an isolated fully renewable energy system integrating batteries and supercapacitors. *Energy Convers Manag* 2021;245:114584. <https://doi.org/10.1016/j.enconman.2021.114584>.
- [86] Ram M, Osorio-Aravena JC, Aghahosseini A, Bogdanov D, Breyer C. Job creation during a climate compliant global energy transition across the power, heat, transport, and desalination sectors by 2050. *Energy* 2022;238:121690. <https://doi.org/10.1016/j.energy.2021.121690>.
- [87] Dufo-l R. Optimisation of PV-wind-diesel-battery stand-alone systems to minimise cost and maximise human development index and job. creation 2016;94:280–93. <https://doi.org/10.1016/j.renene.2016.03.065>.
- [88] Cameron L, Van Der Zwaan B. Employment factors for wind and solar energy technologies: a literature review. *Renew Sustain Energy Rev* 2015;45:160–72. <https://doi.org/10.1016/j.rser.2015.01.001>.
- [89] Elmorshedy MF, Allam SM, Rashad EM. Load voltage control and maximum power extraction of a stand-alone wind-driven PMSG including unbalanced operating conditions. 2016.
- [90] Gonzalez-Espín F, Figueres E, Garcera G, González-Medina R, Pascual M. Measurement of the loop gain frequency response of digitally controlled power converters. *IEEE Trans Ind Electron* 2010;57:2785–96. <https://doi.org/10.1109/TIE.2009.2036635>.
- [91] Services Ain Sokhna industrial area-1. <https://sczone.eg/services/ain-sokhna-industrial-area-1/>.
- [92] Egypt Interest Rate. Data - 1991-2021 historical - 2023 forecast - calendar 2022. <https://tradingeconomics.com/egypt/interest-rate>; 2022.